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# Understanding Over-Optimism in Tax Revenue-to-GDP Forecasts under IMF-supported Programs

Jean-Marc Atsebi, Qiaoe Chen, Nikolay Gueorguiev, Magali Pinat

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Fiscal Affairs Department

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**Prepared by Jean-Marc Atsebi, Qiaoe Chen, Nikolay Gueorguiev, Magali Pinat***

Authorized for distribution by S. Ali Abbas
October 2026

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**ABSTRACT:** Revenue mobilization is a key objective in many IMF-supported programs, yet tax revenue-to-GDP forecasts often fall short of outcomes. This paper analyzes the underlying drivers of these forecast misses by leveraging two novel global datasets on tax policy yields and tax administration capacity. It finds that tax revenue-to-GDP forecast errors are primarily driven by over-optimism in forecasted yields of tax policy measures and tax buoyancy assumptions, as well as macroeconomic forecast errors, particularly for growth and trade openness. Stronger tax administration capacity and well-implemented and high-quality revenue-related structural reforms are associated with lower forecast errors. Additional empirical analysis shows that over-optimism in forecasted tax policy yields declines significantly when revenue measures are well-designed, part of large-measure revenue packages, and accompanied by high-quality revenue-related structural reforms. Improving forecast realism requires better-grounded yield estimations and tax buoyancy assumptions, explicit consideration of implementation capacity, and a focus on well-designed revenue-related structural reforms.

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| Author's E-Mail Address: | JAtsebi@IMF.org; QChen@IMF.org; NGueorguiev@IMF.org; MPinat@IMF.org |

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WORKING PAPERS

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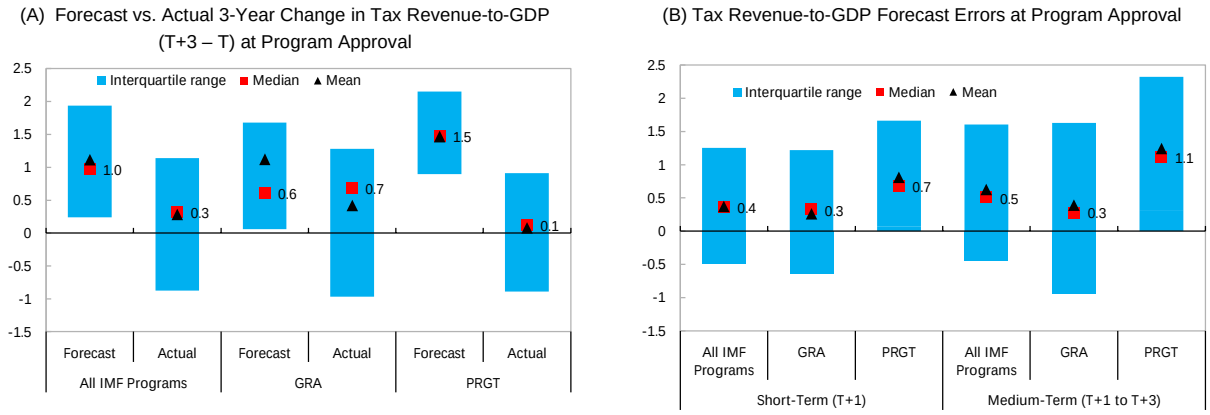
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# I. Introduction

Over-optimistic tax revenue-to-GDP forecasts carry real costs.[1] When revenues underperform expectations, fiscal targets may be missed—eroding fiscal buffers and heightening vulnerability—or met only through abrupt expenditure cuts that disproportionately affect growth-enhancing investment and weaken medium-term growth prospects (Golosov, 2002; Danninger, 2005; Frankel, 2011; Cangiano and Pathak, 2019). Persistent tax revenue-to-GDP shortfalls can further distort incentives by placing excessive pressure on taxpayers to meet unrealistic targets, while exceptional revenues in one period may induce authorities to defer collections to sustain ambitious targets in subsequent periods (Tang et al., 2025). This is particularly important for countries under IMF-supported programs, where tax revenue-to-GDP projections play a central role in anchoring fiscal frameworks and conditionality, and where over-optimism can lead to overly ambitious adjustment paths, repeated program recalibration, and difficult trade-offs between fiscal credibility, growth, and social objectives.

**Figure 1. Tax Revenue-to-GDP in IMF Programs: Projections and Outcomes, 2015–24**




Sources: IMF WEO, IMF staff calculations.
Notes: The charts report the interquartile range (blue bars), mean (black triangles), and median (red squares) of forecasted versus actual changes in tax revenue-to-GDP (Panel A) and forecast errors (Panel B) for 99 completed IMF-supported programs (2015–2024) across 66 countries. Panel A compares forecasted and actual changes between the approval year and three years later. Panel B defines forecast errors as the difference between projections at approval and outturns (October 2025 WEO), for the short-term (first program year) and medium-term (three-year post-approval average).

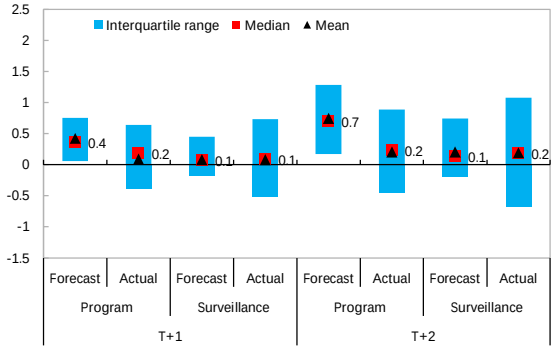
Figure 1 illustrates that tax revenue-to-GDP tends to be over-optimistic in IMF programs.[2] Panel (A) shows that IMF-supported programs forecasted a median increase in tax revenue-to-GDP of 1 percentage points by the third year, but the median actual tax revenue increase was only 0.3 percentage point. Forecasts are higher and outturns lower in Poverty Reduction and Growth Trust (PRGT)-supported programs than in General Resource Account (GRA)-supported programs. Panel (B) shows that both the mean and median forecast errors are positive, with actual tax revenue-to-GDP below projected levels by 0.4–0.5 percentage point in the first year

[1] This paper focuses on the tax revenue-to-GDP ratio rather than nominal tax revenue, as ratios enable meaningful cross-country comparisons and policy-relevant analysis of the revenue forecast errors, which is in line with the literature on fiscal forecast errors. Thus, an adverse growth surprise that lowers nominal revenues but leaves the revenue/GDP ratio unchanged—given unit elasticity––will not be captured by this analysis, even though such a surprise may increase the nominal deficit.

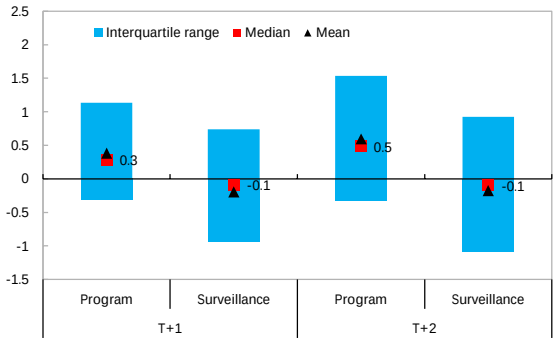
[2] These findings remain robust even when excluding the COVID-19 pandemic years 2020–21.

## Figure 2. Tax Revenue-to-GDP in Programs and Surveillance: Projections and Outcomes, 2015–24

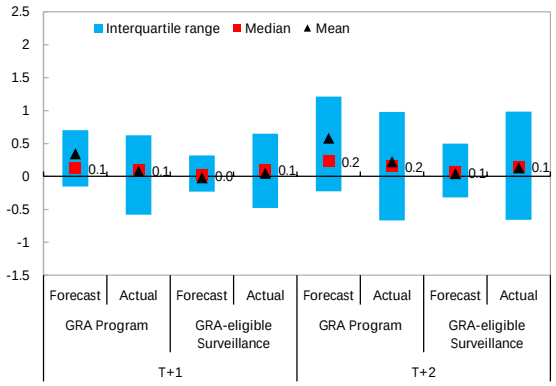
(A) All Countries. Forecast vs. Actual Change in Tax Revenue-to-GDP in IMF-supported Programs and Surveillance



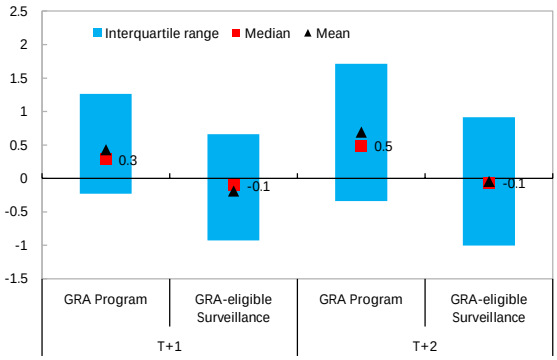

(B) All Countries. Tax Revenue-to-GDP Forecast Errors in IMF-supported Programs and Surveillance



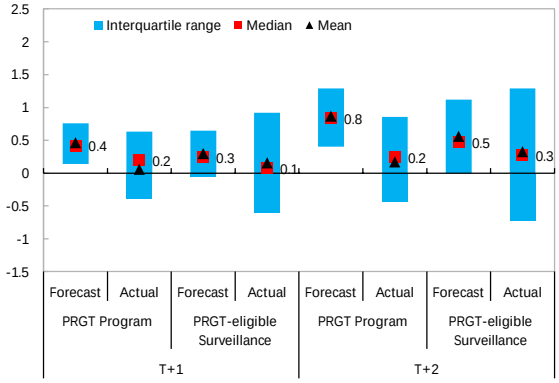

(C) GRA-eligible Countries. Forecast vs. Actual Change in Tax Revenue-to-GDP in IMF-supported Programs and Surveillance



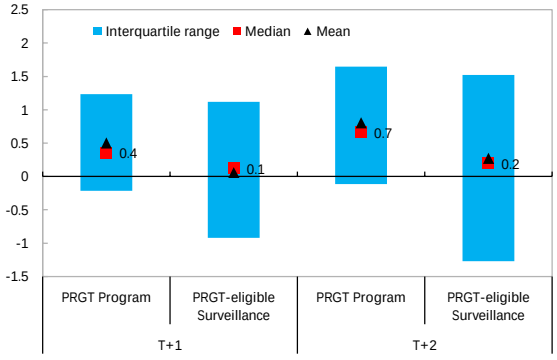

(D) GRA-eligible Countries. Tax Revenue-to-GDP Forecast Errors in IMF-supported Programs and Surveillance




(E) PRGT-eligible Countries. Forecast vs. Actual Change in Tax Revenue-to-GDP in IMF-supported Programs and Surveillance




(F) PRGT-eligible Countries. Tax Revenue-to-GDP Forecast Errors in IMF-supported Programs and Surveillance




Sources: IMF WEO, IMF staff calculations.
Notes: The charts report the interquartile range (blue bars), mean (black triangles), and median (red squares) of changes in tax revenue (Panels A, C, and E) and tax revenue-to-GDP forecast errors (Panels B, D and F). The sample covers 179 countries and 99 completed IMF-supported programs (2015–2024). Panels A, C, and E show the distribution of one- and two-year-ahead changes in forecasted and actual tax revenue-to-GDP as reported in October WEO vintages and incorporates revisions by including projection at program reviews. Panels B, D, and F show the distribution of one- and two-year-ahead tax revenue-to-GDP forecast errors.

and over the three years following program approval. Shortfalls are more pronounced in PRGT-supported programs, where they stand at 1.1 percentage point of GDP over the three years after approval. The interquartile ranges highlight considerable heterogeneity across programs, including a notable negative forecast error segment (signaling more conservative forecasts) in GRA-supported programs.

Figure 2 compares forecasted and actual changes in tax revenue-to-GDP, as well as tax revenue-to-GDP forecast errors, across IMF-supported programs and surveillance countries over 2015–24.[3] The comparison is shown separately for all countries, GRA-eligible countries, and PRGT-eligible countries. Panel (A), (C), and (E) show that forecasted changes in tax revenue-to-GDP are notably higher for countries under IMF-supported programs than for those under surveillance, yet actual changes are only modestly higher (in some cases similar) in program countries. Zooming in, Panel (B), (D), and (F) display larger tax revenue-to-GDP forecast errors in program countries, with distributions more skewed toward positive values, indicating greater overprediction of tax revenue under programs. These findings are especially striking for PRGT-supported programs. The median forecast errors ranges between 0.3 and 0.5 percentage points of GDP in GRA-supported programs, and 0.4 and 0.7 percentage points of GDP in PRGT-supported programs. By contrast, forecasts in surveillance countries remain broadly aligned with outcomes, despite substantial dispersion.

Against this backdrop, this paper seeks to identify the underlying drivers of over-optimism in tax revenue-to-GDP forecasts made by IMF teams as well as the determinants of its variation by country groups. Forecast errors may arise from overly optimistic projections of yields from tax policy measures, overestimated tax buoyancy assumptions (how much revenues rise with the rise of the tax base), macroeconomic forecast errors (including due to unforeseen shocks), or implementation shortfalls. Institutional factors—such as tax administration capacity or the depth (reach and quality) of revenue-related structural reforms—may also influence the extent to which projected tax revenue gains materialize. Ascertaining the role and relative importance of these factors will help IMF country teams improve their tax revenue-to-GDP forecasts and raise the quality of the Fund's program and surveillance work.

Our paper relates to several strands of literature:

- First, a large body of work examines the accuracy and optimism of macroeconomic forecasts, including those produced by the IMF and other institutions (Timmermann, 2007; Eicher et al., 2019; Carrière-Swallow and Marzluf, 2022; Estefania-Flores et al., 2023). Several studies document optimism in macroeconomic projections, particularly in growth forecasts during adjustment programs (Blanchard and Leigh, 2013; Kim et al., 2021; Carrière-Swallow and Marzluf, 2022). Earlier work focusing on IMF-supported programs has found evidence of optimistic fiscal projections, particularly in low-income countries and in programs with ambitious policy adjustments (Golosov and King, 2002; Musso and Phillips, 2001; Ho and Mauro, 2016; Ismail et al., 2020).

- Second, a related literature analyzes fiscal forecast accuracy, mostly using national government forecasts. Studies on European countries under the Stability and Growth Pact find that budget balances and revenues often fall short of official projections (Beetsma et al., 2009; Brück and Stephan, 2006; Marinheiro, 2010). Similar patterns have been documented in Latin America and other regions (Frankel, 2011; Hadzi-

[3] In contrast to Figure 1, the program-related forecasts in Figure 2 also incorporate intra-program forecasts made at program reviews. This is necessary to ensure comparability with forecasts in surveillance countries, which are updated at least annually.

Vaskov et al., 2021). These studies typically attribute fiscal forecast errors to optimistic macroeconomic assumptions, institutional weaknesses, and political incentives.

- Third, a growing literature focuses specifically on tax revenue forecasting. Previous studies show that forecast errors can arise from incorrect macroeconomic assumptions, misestimated elasticities, institutional constraints, or political economy considerations (Danninger, 2005; Danninger and Kyobe, 2005a, 2025b; Jochimsen and Lehmann, 2017; Göttert and Lehmann, 2021). However, relatively little research has examined the drivers of tax revenue forecast errors in the context of IMF-supported programs—particularly in recent years—and with explicit attention to both technical factors (such as the realism of expected revenue yields from tax policy measures and assumptions about tax buoyancy) and policy-related ones (e.g., implementation shortfalls, depth and quality of revenue-related structural reforms, and tax administration capacity).

Our analysis makes several contributions to literature. First, to the best of our knowledge, this paper is the first to examine whether over-optimism in the estimation of tax policy yields and weaknesses in tax administration capacity affect the accuracy of tax revenue-to-GDP forecasts. We leverage two novel databases: (i) the Global Tax Policy and Revenue Database (TAPRED)—collecting data on the expected yields of tax policy measures (Atsebi and Goyal, 2026), and (ii) the database on operational strength of tax administrations (Atsebi et al., 2025). Second, we provide new evidence on the role of other factors, including forecast errors in tax buoyancy, macro-economic factors, shortfalls in implementation of revenue policy measures, and quality and implementation of revenue-related structural reforms under IMF-supported programs. Third, we cover the period 2015–2024, which was characterized by renewed emphasis on domestic revenue mobilization. Fourth, we document differences in tax revenue-to-GDP forecast errors between countries under IMF-supported programs and surveillance countries.

Our results show that tax revenue-to-GDP forecast errors in IMF-supported programs are closely linked to tax policy yield estimates, tax buoyancy assumptions, tax administration capacity, and macroeconomic forecast errors.

- First, tax policy yield estimates under IMF-supported programs appear over-optimistic ex-ante; controlling for policy implementation shortfalls reduces (but does not fully eliminate) the related forecast error.

- Second, when revenue-related structural reforms under IMF-supported programs are high-quality and fully implemented, over-optimism in tax policy yields is effectively eliminated.[4]

- Third, optimistic tax buoyancy assumptions also play a central role in explaining the forecast errors; addressing this seems a low-hanging fruit.

- Fourth, stronger tax administration capacity significantly reduces the forecast errors.

- Fifth, macroeconomic forecast errors, particularly those for growth and trade openness, also contribute to the tax revenue-to-GDP forecast errors.

[4] Indeed, when revenue-related structural reforms are both fully implemented and of higher depth, the conditional effect of tax policy yields on forecast errors is no longer statistically distinguishable from zero.

- Sixth, more ambitious revenue targets are associated with larger target misses; however, the main drivers of forecast errors are primarily related to other factors—assumptions embedded in revenue projections, quality of measures, implementation gaps, and tax administration capacity.

- Seventh, the analysis of the materialization of projected ex-ante revenue gains from tax policy measures indicates that, on average, only 40 percent of them materialize under IMF-supported programs, though the materialization rate increases substantially (to 95 percent) when revenue-related structural reforms in these programs are of high quality and fully implemented.

- Eighth, this analysis also shows that while IMF-supported programs embed greater revenue ambition as part of an intended fiscal regime shift—the expected revenue yields are 1.6 times higher than in surveillance—this ambition also delivers higher realized tax policy gains than surveillance cases.

- Finally, only large tax policy measures deliver statistically significant revenue gains, while fragmented bundles of small measures have no discernible revenue effect and tend to generate over-optimism by overstating their projected yields in contrast to near-zero realized outcomes.

The remainder of the paper is organized as follows. Section II describes the data. Section III introduces the empirical methodology. Section IV presents the baseline results, robustness checks, and sensitivity analyses. Section V concludes and discusses policy implications.

# II. Data

Our sample includes IMF-supported programs that started and ended between 2015 and 2024, when the world faced major global shocks—most notably the COVID-19 pandemic—that put revenue projections and fiscal frameworks under significant stress. Examining this recent decade, therefore, provides a stringent test of whether revenue projections display significant systematic optimism in contemporary IMF-supported programs. During this period, there were 99 completed IMF-supported programs, of which 31 disbursing GRA programs, 41 PRGT programs (including blended programs), and 27 non-disbursing programs. The description of IMF-supported programs and information on their design, including revenue-related structural reforms, is drawn from the IMF's Monitoring of Fund Arrangements (MONA) database. We also use indicators capturing the depth and implementation of revenue-related structural reforms, which measure the extent to which programs incorporate substantive revenue reforms involving durable institutional changes, rather than purely administrative or short-term policy adjustments, from the IMF Review of Conditionality (IMF, 2018 a).[5]

The novel TAPRED, constructed by Atsebi and Goyal (2026), provides information on the expected revenue yields per year from tax policy measures and their implementation status.[6] This allows us to test whether over-optimism in tax policy yield estimation contributes to over-optimistic tax revenue-to-GDP forecasts. Specifically,

[5] Structural reforms are classified into measures of high, medium, or low depth, where high depth are reforms that lead to permanent institutional changes, such as by involving legislative changes (parliamentary approval), or reforms with long-lasting impact (e.g., civil service reforms, SOE reforms, privatization); medium depth are reforms that lead to a significant change but are one-off in nature (e.g., budget approval or one-time change in tariff rates as compared to a permanent change such as institutionalizing an automatic tariff adjustment mechanism), and low depth are reforms that in themselves do not bring about change but are steps towards a change.

[6] Please see Atsebi and Goyal (2026) for a description of the database, as well as new stylized facts on tax policy measures.

TAPRED covers 2,360 tax policy measures and their associated yields estimation over 2015–2024.[7] Using this dataset, we manually validated the implementation status and the timing (year of implementation/impact variables) ex-post using non-IMF sources (e.g., budget documents, official announcements, and media reports) and with the support of LLMs. Overall, 71 percent of quantified tax policy measures were fully implemented, 14 percent were implemented with modifications, and 15 percent were not implemented, with similar implementation rates in program and surveillance cases.

The Operational Strength Index (OSI), constructed by Atsebi et al. (2025), measures the strength of tax administration. It is constructed using data from the International Survey on Revenue Administration (ISORA) and captures both the operational practices and institutional foundations of tax administrations. It is based on nine sub-indices reflecting key dimensions of tax administration performance.[8] Higher OSI values reflect stronger administrative capacity.[9]

The World Economic Outlook (WEO) vintages are used to construct the tax revenue-to-GDP and macroeconomic forecast errors, with the October 2025 WEO being used to construct outturns. To capture the forecasts made at program approval ($h = 0$), we use the April or October WEO vintages following the approval date of the program. To build the panel of h-year-ahead forecast errors, we use only the October WEO vintages.[10] We compute forecast errors (forecast minus actual) in tax revenue-to-GDP , tax buoyancy—a measure of tax revenue responsiveness to economic activity and discretionary tax policy—as well as in key macroeconomic variables, including growth, inflation, and trade openness. To account for data revisions and GDP rebasing, we adjust the forecast errors (when expressed in percentage of GDP) by subtracting the average discrepancy in forecast errors observed over the three years preceding program approval or the relevant forecast year. This correction removes systematic differences arising from data revisions and GDP rebasing rather than forecast performance.[11]

Data on fiscal councils and fiscal rules come from the IMF Fiscal Rules Dataset, while institutional quality variables are drawn from the Worldwide Governance Indicators (WGI). Appendix Table A.I.1 presents a description of the variable used in this paper, Appendix Table A.I.2 and A.I.4 provide summary statistics, and Appendix Table A.I.3 and A.I.5 provide correlations between variables.

[7] The Global Tax Policy and Revenue Evaluation Database (TAPRED) is the most comprehensive narrative database that systematically records tax policy measures and their associated revenue yields across advanced economies and, critically, emerging and developing countries. It extracts 3,922 measures with quantified revenue yields from IMF SRs and MEFPs and covers 193 countries and territories, including 42 advanced economies, 93 emerging markets, and 58 low-income countries, over the period 1997–2024.

[8] See Atsebi et al. (2025) for more details.

[9] Given that the OSI is available only for the years 2014, 2015, 2016, 2017, and 2022, we use an interpolation to fill the missing values. Restricting the sample to years with observed OSI values would substantially reduce the scope of the analysis, particularly after 2017, when OSI data become sparser, and for coverage of the three years following program approval. This data limitation also prevents us from analyzing how changes in OSI over time affect tax-revenue-to-GDP forecast errors.

[10] Since the April or October vintage used to measure initial forecasts is fixed regardless of when in the year measures were implemented, the effective information window varies across observations. As month-level implementation data are unavailable, we cannot test whether this timing variation systematically affects forecast errors, and flag it as a caveat.

[11] In practice, absent GDP rebasing or data revisions, forecast errors for realized outcomes in years preceding the program (e.g., H-1, H-2 or H-3) should be close to zero, as these values are already observed. However, differences may arise due to subsequent revisions to GDP or fiscal aggregates, especially GDP rebasing. The adjustment therefore uses the average pre-program discrepancy as a proxy for revision effects and nets it out from post-program forecast errors.

# III. Empirical Methodology

Our empirical analyses rely on two complementary approaches to assess the potential drivers of over-optimism in tax revenue-to-GDP forecast: (i) a forecast-error approach, and (ii) a revenue-gain approach. The former examines tax revenue-to-GDP forecast errors and their drivers directly. The latter focuses on expected revenue yields from tax policy measures, evaluates whether they translate into realized increases in tax revenues, and whether systematic over-optimism is embedded in the estimation of tax policy yields themselves. We examine the role of several drivers, including over-optimism in tax policy yield estimates and tax buoyancy assumptions, macroeconomic shocks, implementation shortfalls, low quality of measures, and weaknesses in tax administration.

## A. Forecast-Error Approach

Following Blanchard and Leigh (2013), we investigate whether tax revenue-to-GDP forecast errors have been systematically related to specific drivers. We define the tax revenue-to-GDP forecast error $\widehat{rev}_{c,v,t,h}$ at horizon $h$ as the difference between the $h$-year-ahead forecast—using data in October vintages of the WEO or semiannual IMF forecasts from successive WEO vintages closest to the approval date of the IMF-supported programs—and the corresponding outturn:

$$\widehat{rev}_{c,v,t,h} = E_{v,t-h}[rev_{c,v,t}] - rev_{c,t}, \text{ with } h=1,2,3 \quad (1)$$

where $E_{v,t-h}[rev_{c,v,t}]$ is the expected tax revenue-to-GDP based on the information set available to forecasters at date $t-h$ in vintage $v$ of the WEO, and $rev_{c,t}$ is the outturn in country $c$ and year $t$, based on the October 2025 vintage of the WEO. A positive value of $\widehat{rev}_{c,v,t,h}$ denotes an optimistic tax revenue-to-GDP forecast. We regress the tax revenue-to-GDP forecast error on a set of plausible explanatory variables. The equation estimations are specified as follows:

$$\widehat{rev}_{c,v,t,h} = \alpha + \beta_1 Yield_{c,t} + \beta_2 IQ_{c,t} + \beta_3 Yield_{c,t} XIQ_{c,t} + \beta_4 \widehat{buo}_{c,v,t,h} + \beta_5 \widehat{macro}_{c,v,t,h} + \beta_6 OSI_{c,t-h} + \beta_7 X_{c,t-h} + \beta_8 Z_{c,t} + \upsilon_t + \varepsilon_{c,v,t,h} \quad (2)$$

$$\widehat{rev}_{c,v,t,h} = \alpha + \beta_1 IYield_{c,t} + \beta_2 IModYield_{c,t} + \beta_3 NotIYield_{c,t} + \beta_4 \widehat{buo}_{c,v,t,h} + \beta_5 \widehat{macro}_{c,v,t,h} + \beta_6 X_{c,t-h} + \beta_7 Z_{c,t} + \upsilon_t + \varepsilon_{c,v,t,h} \quad (3)$$

where:

- $Yield_{c,t}$ is the expected revenue yield from tax policy measures for country $c$ in year $t$, from TAPRED. We include it to test whether forecasted revenue gains systematically translate into optimistic tax revenue-to-GDP forecasts. If yields are accurately estimated, the coefficient $\beta_1$ should be zero or not statistically significant; a positive and statistically significant $\beta_1$ would indicate the presence of a systematic bias in yield forecasts, contributing to tax revenue-to-GDP forecast errors. In Eq. (3), this variable is further decomposed by implementation status, where $IYield_{c,t}$ denotes the yield of fully implemented measures, $IModYield_{c,t}$ is the yield of measures implemented with modifications, and $NotIYield_{c,t}$ is the yield of non-implemented measures. Our expectation is that measures that are only partially implemented or not implemented at all should be associated with larger tax revenue-to-GDP forecast errors.

- $IQ_{c,t}$ is a composite index ranging from zero to one that captures both the implementation and depth of revenue-related structural reforms in IMF-supported programs; higher value indicates higher implementation and/or depth of revenue-related structural reforms.[12] Its inclusion is central to the analysis because, in program settings, revenue outcomes depend not only on announced tax measures, but also on whether supporting reforms—such as strategies, legislation, regulations, or administrative actions—have higher depth and are implemented on time. Intuitively, higher $IQ_{c,t}$ should increase the likelihood that expected yields are realized and, therefore, reduce optimistic tax revenue-to-GDP forecast errors.
- $\widehat{buo}_{c,v,t,h}$ is the tax buoyancy forecast error, where tax buoyancy is measured by the change in nominal tax revenue to the change in nominal GDP assumed in IMF forecasts and captures (i) the responsiveness of tax to the change in the tax base and (ii) discretionary policy (tax policy reforms, tax administration reforms). We include it to control for broader forecast errors in the assumed revenue-to-base relationship, which is important because tax revenue-to-GDP forecast errors may reflect not only misestimated tax policy yields, but also incorrect assumptions about how revenues respond to macroeconomic developments and changes in the tax base.[13] A positive and statistically significant coefficient would indicate that overly optimistic assumptions about tax buoyancy are associated with larger optimistic tax revenue-to-GDP forecast errors, whereas a coefficient close to zero would suggest that buoyancy assumptions do not systematically contribute to forecast errors once expected tax policy yields are controlled for.
- $\widehat{macro}_{c,v,t,h}$ is a set of forecast errors for growth, inflation, and trade openness.[14] These variables are included to control for macroeconomic forecast errors that can mechanically affect tax revenue projections through their impact on tax bases. Growth forecast errors capture deviations in economic activity affecting income, profits, and consumption taxes; inflation forecast errors capture deviations in nominal tax bases and collection dynamics; and trade openness forecast errors capture deviations in imports and exports, which are particularly important for trade taxes and VAT on imports. Positive forecast errors in these variables are expected to be associated with positive tax revenue-to-GDP forecast errors, as stronger-than-expected macroeconomic conditions generally increase tax collection relative to initial projections.
- $OSI_{c,t-h}$ captures the strength of tax administration. It is included to control for differences in institutional and administrative capacity that affect the ability of countries to collect projected revenues and implement tax measures effectively. Stronger tax administrations are expected to improve compliance, enforcement, filing and payment systems, and the translation of economic activity into actual revenue collection. Therefore, higher values of $OSI_{c,t-h}$ should be associated with smaller tax

[12] Implementation and depth are combined in one index, as it is their combined effects that matter. Deep reforms that are not implemented do not deliver results, while reforms that are implemented but shallow are unlikely to generate meaningful gains.
[13] Because the tax policy yields variable is explicitly included in the regression alongside the tax buoyancy variable, the coefficient associated with tax buoyancy captures factors other than tax policy measures, including tax elasticity and taxpayers' behavioral changes. We also tested an alternative approach that subtracts tax policy yields from tax buoyancy, including adjustments to both forecasted and realized buoyancy. The main findings are broadly robust.
[14] While tax revenues are generally more closely linked to nominal GDP, our dependent variable is the tax-revenue-to-GDP forecast error, where GDP also enters the denominator. We therefore use separate growth and inflation forecast errors rather than nominal GDP forecast errors to disentangle the channels through which macroeconomic misperceptions affect tax revenue-to-GDP forecast errors. Real growth errors capture misjudgment of underlying economic activity and tax bases, while inflation errors capture price dynamics that may affect revenues and GDP differently. Using nominal GDP forecast errors alone would combine these effects and make interpretation less informative.

revenue-to-GDP forecast errors, implying a negative coefficient if stronger tax administration improves tax revenue forecast realism.

- $X_{c,t-h}$ captures program design at approval through variables such as forecasted three-year change in tax revenue-to-GDP, debt, primary balance, and forecasted average growth over three years. These variables are included to control for the ambition and macro-fiscal context of the program, since more ambitious adjustment paths may be more prone to optimistic revenue projections. A positive coefficient would suggest that stronger programmed improvements are associated with larger optimistic tax revenue-to-GDP forecast errors.
- $Z_{c,t}$ captures the presence of fiscal rules and council, as well as quality of institutions (e.g. regulatory quality, control of corruption, etc.). These controls account for differences in fiscal discipline, credibility, transparency, and governance that may affect the realism of tax revenue-to-GDP forecasts and the capacity to deliver projected tax gains. Stronger fiscal institutions and better governance are expected to reduce optimistic tax revenue-to-GDP forecast errors, implying a negative coefficient.
- $\alpha$ captures the average unexplained component of tax revenue-to-GDP forecast errors. A positive and statistically significant value indicates that actual revenue outturns tend to be lower than projected, after accounting for observable explanatory drivers.
- $\upsilon_t$ are year dummies, capturing common shocks (e.g., COVID-19 and 2014-16 oil shock).

To sum up, if none of these plausible determinants are correlated with tax-to-GDP forecast errors, then all slope coefficients satisfy $\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = \beta_8 = 0$ in terms of statistical significance. Moreover, in the absence of unexplained factor driving tax-to-GDP forecast errors, the intercept term satisfies $\alpha = 0$, and if common shocks have no impact, the time effects satisfy $\upsilon_t = 0$.

The forecast-error approach is applied to two distinct samples. The first consists of tax-to-GDP forecast errors measured at the time of program approval. The second comprises a panel of *h*-year-ahead forecast errors drawn from both Fund-supported programs—including program reviews—and surveillance.[15] The second sample serves to assess the robustness of the results by evaluating whether the patterns identified at program approval persist when reviews are incorporated, while also permitting benchmarking against surveillance projections.

## B. Revenue-Gain Approach

The revenue-gain approach relates changes in the tax revenue-to-GDP ratio to the expected yield of tax policy measures and a set of controls capturing macroeconomic conditions and institutional factors. This specification provides a complementary analysis to the forecast-error approach, allowing us to assess the robustness of our findings by examining whether revenue projections embedded in tax policy measures exhibit systematic over-optimism. In addition, it enables us to evaluate whether IMF-supported programs are associated with more effective tax policy measures in raising revenue relative to surveillance countries. The estimated equation is specified as follows:

[15] It is worth noting that this comparison between program and surveillance cases may be influenced by selection effects: countries entering IMF-supported programs tend to face more difficult macroeconomic and institutional conditions, which may independently contribute to larger forecast errors. The observed differences across groups should therefore be interpreted cautiously as reflecting a combination of program-specific effects and underlying country characteristics rather than the effect of program status per se.

$$\Delta rev_{c,t} = \alpha + \theta_1 Yield_{c,t} + \theta_2 X_{c,t} + \mu_c + \lambda_t + \varepsilon_{c,t} \quad (4)$$

where $\Delta rev_{c,t}$denotes the change in the tax revenue-to-GDP ratio for country $c$ in year $t$. $X_{c,t}$ is a vector of control variables capturing macroeconomic conditions and structural characteristics identified in the literature as key determinants of tax revenue performance (Tanzi, 1992; Gupta, 2007; Mitra, 2017; IMF, 2015, 2018 b, 2019; Adan et al., 2023; Atsebi et al., 2025). These controls include real GDP growth, inflation, change in terms of trade, trade openness, external debt, oil exports, and agricultural value added. Country fixed effects $\mu_c$ account for time-invariant country-specific heterogeneity, while time fixed effects $\lambda_t$ capture global shocks and common trends affecting all countries. $\varepsilon_{c,t}$ denotes the error term.

The coefficient $\theta_1$ captures the extent to which projected revenue gains from tax policy measures translate into realized increases in tax revenue. A value of $\theta_1 = 1$ implies full materialization of projected yields, indicating the absence of systematic over-optimism in the revenue yields from tax policy measures (conditional on full implementation, see below). By contrast, a coefficient significantly below unity ($\theta_1 < 1$) suggests that projected yields tend to overstate realized revenue gains, reflecting potential over-optimism in the revenue yields from tax policy measures or incomplete implementation. The magnitude of the deviation from unity therefore provides a measure of the extent of optimism.

Consistent with the forecast-error approach, we further distinguish tax policy measures by implementation status and interact tax policy yields with the quality of structural reforms in IMF-supported programs. This specification allows us to examine whether implementation gaps and the quality of reforms design contribute to potential over-optimism in projected tax policy yields, and to identify the areas where stronger policy design and implementation could enhance the effectiveness of tax policy in mobilizing revenue.

Finally, we also analyze the quality of tax policy measures by differentiating between *large* and *small* measures based on the size of the revenue yields.[16] Tax policy measures are classified as *large* when the estimated ex-ante yield of an individual measure exceeds 0.05 percent of GDP, and as *small* otherwise. Measures in each category are then aggregated at the country–year level, allowing us to separately compute the total revenue impact of *large* and *small* reforms. This classification enables an assessment of whether the bundling of numerous small measures systematically generates economically meaningful revenue gains.

# IV. Baseline Findings

This section presents the baseline results, shedding light on the drivers of tax revenue-to-GDP forecast errors and zooming in on tax policy yield estimates in the revenue gain approach. It also assesses the extent to which IMF-supported programs are associated with more effective tax policy measures in raising revenue, relative to surveillance.

[16] This variable is excluded from the forecast-error approach because larger packages mechanically create more scope for forecast errors, while smaller packages limit that scope. Controlling for size would therefore absorb part of the variation of interest.

# A. Forecast-Error Approach

## 1. Tax Revenue-to-GDP Forecast Errors at Program Approval

Table 1 and Figure 3 report the baseline regression results examining the determinants of tax revenue-to-GDP forecast errors at program approval for different horizons, in line with Equations (2) and (3). Columns (1)–(3) of Table 1 present results for the first year following the program approval ($h = 1$), while columns (4)–(6) report results for the average forecast error in the first three years following program approval ($h = 1, 2, 3$). The estimates show that forecast errors are strongly associated with over-optimistic tax policy yields estimates and tax buoyancy assumptions, as well as with macroeconomic forecast errors in growth and trade openness. Forecast errors are also more prevalent in countries with weaker tax administration capacity, and when there are implementation shortfalls or a lower depth of revenue-related structural reforms.

Specifically, first, tax revenue-to-GDP forecasts are associated with over-optimism in tax policy yield estimates. Based on the baseline specification of Equation (2), a one percentage point of GDP increase in the estimated yield of tax policy measures is associated with a 0.79 percentage point increase in the tax revenue-to-GDP forecast error in the first program year (column (1)) or a 0.49 percentage point over the three years (column (4)). These results suggest that part of the projected revenue gains from tax policy measures materialize more slowly than anticipated. Notably, even fully implemented measures are associated with positive forecast errors—around 0.86 percentage point of GDP in the first year and 0.39 percentage point of GDP over the three years—indicating that revenue estimates tend to be optimistic even when reforms are implemented as planned (columns (3) and (6)). The coefficients associated with modified and non-implemented measures are also positive, showing that forecast errors are generally larger when implementation falls short. Overall, the evidence suggests that both channels are at play: tax policy yields tend to be overestimated even when measures are implemented, while revenue shortfalls are further amplified when measures are weakly designed, modified, or not implemented at all.

Second, stronger implementation and higher depth of revenue-related structural reforms mitigate over-optimism in revenue yield forecasts of tax policy measures. In columns (2) and (5), tax policy yields and the index measuring the implementation and depth of revenue-related structural reforms in IMF-supported programs are interacted. The negative and statistically significant interaction term suggests that higher-quality and better-implemented revenue-related structural reforms reduce the contribution of tax policy yield estimates to forecast errors. An important implication of this interaction is that full implementation alone does not appear sufficient to eliminate over-optimism; rather, it is the combination of full implementation and higher depth or quality reforms that together drive this reduction. Indeed, when revenue-related structural reforms are both fully implemented and of higher depth, the conditional effect of tax policy yields on forecast errors is no longer statistically distinguishable from zero, suggesting that over-optimism is substantially reduced (see the line *"Effect of TP Yield Conditional on Revenue SCs (at Full)"* and its p-value).

Third, the results confirm the central role of tax buoyancy assumptions in explaining forecast errors. Across all specifications, the coefficient on tax buoyancy forecast errors is positive and highly statistically significant. This indicates that when projected tax buoyancy exceeds realized buoyancy, revenue projections tend to be overly optimistic.

**Table 1. Drivers of Tax Revenue-to-GDP Forecast Errors at Program Approval**

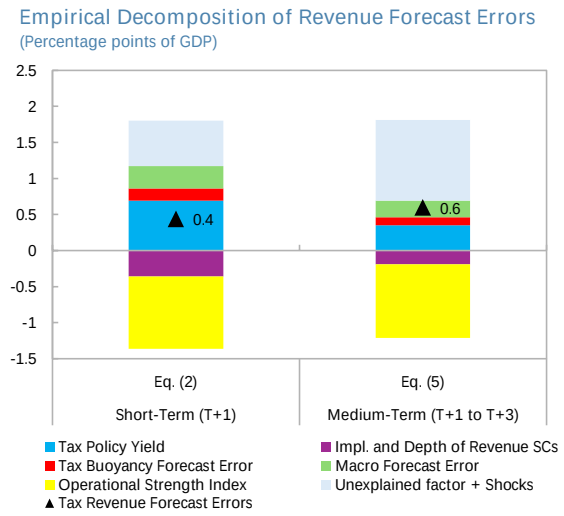
| | (1) h=1 | (2) h=1 | (3) h=1 | (4) h=1,2,3 | (5) h=1,2,3 | (6) h=1,2,3 |
|---|---|---|---|---|---|---|
| Tax Policy Yield | 0.791*** | 1.895*** | | 0.488** | 1.194*** | |
| | (0.244) | (0.554) | | (0.185) | (0.262) | |
| Implementation and Depth of Revenue SCs | | -5.238** | | | -2.997 | |
| | | (2.065) | | | (2.835) | |
| TP Yield X Implementation and Depth of Revenue SCs | | -2.237** | | | -1.525*** | |
| | | (0.915) | | | (0.520) | |
| Fully Implemented TP Yield | | | 0.860*** | | | 0.393* |
| | | | (0.248) | | | (0.206) |
| Implemented with Mod TP Yield | | | 0.056 | | | 0.672* |
| | | | (0.472) | | | (0.344) |
| Not Implemented TP Yield | | | 1.078* | | | 0.926* |
| | | | (0.608) | | | (0.500) |
| Tax Buoyancy Forecast Error | 0.460** | 0.552*** | 0.471*** | 0.569*** | 0.567*** | 0.573*** |
| | (0.173) | (0.161) | (0.171) | (0.124) | (0.113) | (0.126) |
| Growth Forecast Error | 0.172*** | 0.189*** | 0.177*** | 0.050* | 0.050* | 0.048 |
| | (0.056) | (0.051) | (0.056) | (0.029) | (0.027) | (0.030) |
| Inflation Forecast Error | 0.005 | -0.010 | -0.001 | -0.033 | -0.025 | -0.035 |
| | (0.036) | (0.038) | (0.037) | (0.033) | (0.034) | (0.033) |
| Trade Openness Forecast Error | 0.030** | 0.024** | 0.032*** | 0.031*** | 0.030*** | 0.031*** |
| | (0.011) | (0.009) | (0.011) | (0.009) | (0.008) | (0.009) |
| Operational Strength Index | -1.986** | -1.910* | -2.094** | -2.800** | -1.933 | -2.716** |
| | (0.784) | (1.065) | (0.819) | (1.243) | (1.221) | (1.228) |
| Constant | 0.751 | 3.748*** | 0.796 | 1.571** | 3.063* | 1.507** |
| | (0.464) | (1.222) | (0.492) | (0.688) | (1.701) | (0.683) |
| Observations | 79 | 79 | 79 | 246 | 246 | 246 |
| Number of countries | 50 | 50 | 50 | 51 | 51 | 51 |
| Number of programs | 79 | 79 | 79 | 84 | 84 | 84 |
| R-squared | 0.531 | 0.583 | 0.544 | 0.375 | 0.396 | 0.379 |
| WEO vintage FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Tax-to-GDP Forecast Errors Over-Optimism | 0.438 | 0.438 | 0.438 | 0.597 | 0.597 | 0.597 |
| Over-Optimism (p-value) | 0.026 | 0.026 | 0.026 | 0.008 | 0.008 | 0.008 |
| Effect of TP Yield Conditional on Revenue SCs (at Mean) | | 0.999 | | | 0.577 | |
| Conditional Effect (at Mean, p-value) | | 0.000 | | | 0.001 | |
| Effect of TP Yield Conditional on Revenue SCs (at Full) | | -0.341 | | | -0.331 | |
| Conditional Effect (at Full, p-value) | | 0.465 | | | 0.359 | |
| *Shapley Relative Explanatory Power* | | | | | | |
| *Over-Optimism in Yield and Tax Buoyancy* | 38.046 | 33.717 | 34.420 | 39.486 | 36.191 | 35.792 |
| TP Yield | 21.685 | 17.690 | | 7.321 | 6.456 | |
| Fully Implemented TP Yield | | | 17.990 | | | 3.698 |
| Tax Buoyancy Forecast Error | 16.361 | 16.026 | 16.430 | 32.164 | 29.735 | 32.094 |
| *Implementation Shortfall and Quality* | | 10.032 | 4.374 | | 5.565 | 4.996 |
| Implementation and Depth of Revenue SCs | | 2.043 | | | 2.743 | |
| Yield X Implementation and Depth of Revenue SCs | | 7.989 | | | 2.823 | |
| Implemented with Mod TP Yield | | | 0.713 | | | 2.025 |
| Not Implemented TP Yield | | | 3.662 | | | 2.970 |
| *Macro Errors* | 46.797 | 41.694 | 45.312 | 32.992 | 30.299 | 33.037 |
| Growth Forecast Error | 18.495 | 18.320 | 17.690 | 6.437 | 6.085 | 6.319 |
| Inflation Forecast Error | 0.308 | 0.303 | 0.417 | 2.577 | 2.052 | 2.658 |
| Trade Openness Forecast Error | 27.994 | 23.071 | 27.205 | 23.977 | 22.163 | 24.060 |
| *Other Factors* | 15.157 | 14.557 | 15.894 | 27.523 | 27.945 | 26.175 |
| Operational Strength Index | 8.007 | 5.915 | 7.953 | 12.464 | 8.448 | 11.974 |
| Cons + WEO vintage FE | 7.150 | 8.642 | 7.940 | 15.059 | 19.497 | 14.202 |

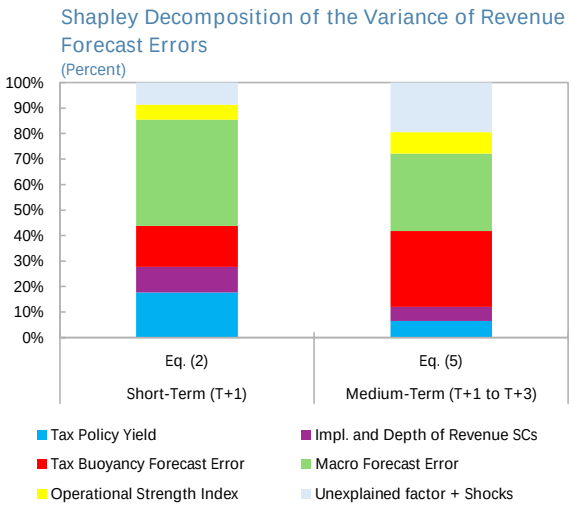
Note: OLS Estimates. Dependent variable is tax revenue-to-GDP forecast errors for the first year following the program approval (columns (1)–(3)) and for the combined horizons covering the first three years following program approval (columns (4)–(6)). In columns (2) and (5), tax policy yields and the index measuring the implementation and depth of revenue-related structural reforms are interacted. Columns (3) and (6) further distinguish tax policy measures by implementation status. Robust standard errors clustered by country are shown in parentheses. The use of *** denotes a coefficient significantly different from zero at the 0.01 confidence level; ** 0.05; * 0.10.

## Figure 3. Empirical Decomposition of the Tax Revenue-to-GDP Forecast Errors and Shapley Decomposition of Their Variance

*Tax revenue-to-GDP forecast errors largely reflect overstated tax policy yields and tax buoyancy, as well as factors unexplained by the model, partly offset by stronger tax administration capacity.*

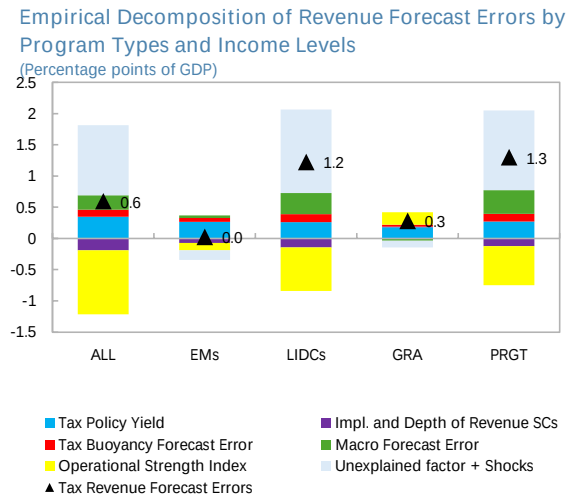
*Their variance is explained by macroeconomic forecast errors, and overstated tax policy yields and tax buoyancy.*

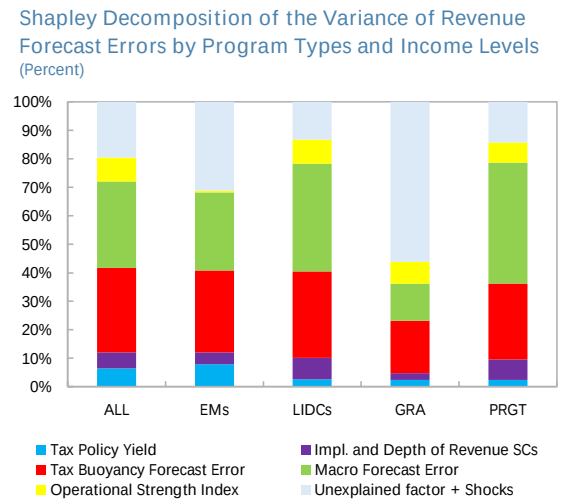



*Overstated tax policy yields and tax buoyancy dominate in EMs and GRA, while LIDCs and PRGT programs are also affected by macroeconomic forecast errors and factors unexplained by the model.*

*There is little change in the drivers of tax revenue-to-GDP forecast error variance.*




Sources: IMF staff calculations.
Note: The empirical decomposition shows the contribution of each explanatory factor, based on the estimated coefficients and the factors' mean values, to the average tax revenue-to-GDP forecast errors. In contrast, the Shapley decomposition explains the contribution of each factor to the overall variation in forecast errors. In short, the first shows what drives the size of errors, while the second shows what explains their dispersion. The upper panel of charts is based on Table 1, while the lower panel of charts is based on Table A.II.4.

Fourth, the strength of tax administration plays an important mitigating role. The coefficients associated with the OSI are negative and statistically significant in most specifications. This suggests that stronger tax administration is associated with smaller tax revenue-to-GDP forecast errors. Quantitatively, the difference in OSI between countries at the 25$^{th}$ and 75$^{th}$ percentiles of the distribution is associated with a smaller forecast error of roughly 0.5 percentage point of GDP, suggesting that IMF teams' tax revenue-to-GDP forecasts are more reliable when the administrative capacity of the country is stronger. This pattern suggests stronger tax administration improves the realism of revenue forecasts by making them more consistent with countries' actual collection capacity.

Fifth, the results confirm that macroeconomic forecast errors contribute to tax revenue-to-GDP forecast errors. In columns (1)–(4), the coefficients on growth forecast errors are positive and statistically significant. A one percentage point overestimation of real GDP growth is associated with an increase in tax revenue-to-GDP forecast errors of roughly 0.17 percentage point of GDP in the first year and around 0.05 percentage point of GDP over the three-year horizon. Similarly, forecast errors in trade openness are positively associated with tax revenue-to-GDP forecast errors across specifications, with a 10-percentage point overestimation raising the tax revenue-to-GDP forecast error by about 0.3 percentage points of GDP. By contrast, inflation forecast errors do not appear to be systematically associated with tax revenue-to-GDP forecast errors.

Finally, the results also point to the presence of additional, unexplained factors driving tax revenue-to-GDP forecast errors. The constant term remains positive and statistically significant in several specifications, indicating that actual revenue outturns tend to be lower than projected even after controlling for the factors discussed above. At the same time, most of the WEO vintage fixed effects are statistically insignificant, suggesting that common global shocks—such as the COVID-19 pandemic or commodity price fluctuations—played a relatively limited role in explaining the observed forecast errors.[17]

The Shapley decomposition reported at the bottom of Table 1 and in Figure 3 (upper panel of charts) further illustrates the relative contribution of these factors in explaining the variance of tax revenue-to-GDP forecast errors. Macroeconomic forecast errors account for a substantial share of the explanatory power, particularly through assumptions related to growth and trade openness. In addition, over-optimism in tax policy yield estimations and tax buoyancy assumptions—captured by tax policy yield estimates and buoyancy assumptions—also explains a large fraction of variance in forecast errors. Finally, implementation shortfalls and the quality of revenue-related structural reforms, as well as the strength of tax administration, also contribute to the variance in forecast errors, although to a lesser extent.

## 2. Panel of Forecast Errors and Benchmarking Against Surveillance

Here we extend the analysis to (i) a larger panel of one- and two-year-ahead tax revenue-to-GDP forecast errors, and (ii) also distinguish between IMF-supported program periods and surveillance. The focus on shorter forecast horizons allows us to incorporate program reviews, which should, in principle, embed updated information and therefore reduce forecast errors.

The results reported in Table 2 broadly corroborate the baseline findings obtained at program approval, with some differences between IMF-supported programs and surveillance. First, tax policy yield estimates remain

[17] A figure plotting tax revenue-to-GDP forecast errors by year shows no statistically significant increase in forecast errors during major global shocks, including the oil price shocks of 2015–16 and 2022–23, as well as the COVID-19 pandemic. This suggests that these episodes did not systematically amplify tax revenue-to-GDP forecast errors.

positively associated with tax revenue-to-GDP forecast errors in IMF-supported programs at both one- and two-year horizons (columns (1)–(4)), while the coefficients on tax policy yield variables are smaller and statistically insignificant in surveillance (columns (7)–(10)).[18] Second, stronger implementation and higher depth of revenue-related structural reforms are associated with a reduction in the contribution of projected tax policy yields to forecast errors, though some of the relevant interaction coefficients are only marginally significant (columns (3)–(4)). Third, the coefficients across implementation categories point broadly in the expected direction—with non-implemented measures tending toward largest forecast errors, while fully implemented measures toward smaller ones (columns (5)–(6)). Notably, non-implemented measures are significantly associated with forecast errors at the one-year horizon. This suggests that implementation gaps become increasingly apparent and consequential as outturns are measured closer in time to the forecast. Fourth, tax buoyancy assumptions continue to play a central role, with buoyancy forecast errors consistently and strongly associated with tax revenue-to-GDP forecast errors across all specifications. Finally, stronger tax administration capacity reduces forecast errors, as captured by the negative and significant coefficient on the OSI, and the positive intercept in program specifications points to the presence of additional factors driving tax-to-GDP forecast errors under IMF-supported programs.

## 3. Aspects of Program Design

The fiscal aspect of program design may influence tax revenue-to-GDP forecast errors because IMF-supported programs effectively amount to a “regime change” for the economy, aiming to achieve ambitious fiscal adjustment targets and debt stabilization path. This means that larger forecast errors in programs may partly reflect higher intended ambition rather than weaker performance per se. Yet this is uncharted territory with significant uncertainty about the economy’s reaction to the new policies. This could potentially amplify tax revenue-to-GDP forecast errors.

Table 3 shows the result by controlling for key program design variables, including the projected change in the tax revenue-to-GDP ratio, the primary fiscal balance, public debt, and average real GDP growth over the three years following program approval, relative to the year prior to the program. While the baseline findings remain largely unchanged, most program design variables—such as projected fiscal consolidation, debt dynamics, and growth assumptions—do not significantly affect tax revenue-to-GDP forecast errors. Still, the positive and significant coefficient on the change in forecasted tax revenue-to-GDP suggests that more ambitious programs in terms of revenue mobilization are associated with greater tax revenue-to-GDP forecast errors. For each 1 percentage point of GDP higher forecasted increase in tax revenue-to-GDP by the third year of the program, the tax revenue-to-GDP forecast error increases by about 0.3 percentage point of GDP.[19] Most importantly, the core coefficients on tax policy yield, tax buoyancy forecast errors, and the interaction with the implementation and depth of revenue-related structural reforms remain stable across specifications. This confirms that the main drivers of tax revenue-to-GDP forecast errors are not primarily related to the broader fiscal framework of program design, but rather to assumptions embedded in revenue projections, quality of measures, implementation gaps, and tax administration capacity.

[18] The lower forecast errors observed in surveillance relative to IMF-supported programs should not be interpreted as evidence of stronger forecasting performance per se. In surveillance contexts, large discretionary revenue increases are rarely contemplated, which mechanically limits the scope for large forecast errors. The program–surveillance contrast therefore reflects differences in ex-ante ambition as much as differences in forecast accuracy.

[19] In a separate analysis (not shown here), we find that larger forecasted revenue increases also tend to result in larger realized revenue increases, though the relationship is far from one-to-one and weakens with the forecast horizon.

## Table 2. Drivers of One- and Two-Year-Ahead Tax Revenue-to-GDP Forecast Errors, IMF-Supported Programs and Surveillance

| | (1) Program (h=1) | (2) Program (h=2) | (3) Program (h=1) | (4) Program (h=2) | (5) Program (h=1) | (6) Program (h=2) | (7) Surveillance (h=1) | (8) Surveillance (h=2) | (9) Surveillance (h=1) | (10) Surveillance (h=2) |
|---|---|---|---|---|---|---|---|---|---|---|
| Tax Policy Yield | 0.332** | 0.343** | 0.404** | 0.615*** | | | 0.109 | 0.252 | | |
| | (0.142) | (0.166) | (0.173) | (0.219) | | | (0.134) | (0.200) | | |
| Implementation and Depth of Revenue SCs | | | -1.674* | -1.672 | | | | | | |
| | | | (0.944) | (1.870) | | | | | | |
| TP Yield X Implementation and Depth of Revenue SCs | | | -0.317 | -0.803* | | | | | | |
| | | | (0.431) | (0.429) | | | | | | |
| Implemented Measures Annual Yield | | | | | 0.232* | 0.331* | | | 0.093 | 0.237 |
| | | | | | (0.135) | (0.178) | | | (0.111) | (0.157) |
| Implemented Mod Measures Annual Yield | | | | | 0.448** | 0.729*** | | | -0.141 | -0.453 |
| | | | | | (0.224) | (0.235) | | | (0.345) | (0.451) |
| Not Implemented Measures Annual Yield | | | | | 0.938*** | 0.304 | | | 0.408 | 0.831 |
| | | | | | (0.226) | (0.282) | | | (0.514) | (0.687) |
| Tax Buoyancy Forecast Error | 0.536*** | 0.337*** | 0.535*** | 0.346*** | 0.553*** | 0.332*** | 0.490*** | 0.426*** | 0.486*** | 0.425*** |
| | (0.096) | (0.090) | (0.097) | (0.091) | (0.097) | (0.091) | (0.065) | (0.067) | (0.066) | (0.068) |
| Growth Forecast Error | 0.038 | 0.021 | 0.040 | 0.026 | 0.037 | 0.021 | 0.026 | 0.033 | 0.025 | 0.034 |
| | (0.024) | (0.028) | (0.024) | (0.029) | (0.024) | (0.028) | (0.034) | (0.032) | (0.034) | (0.031) |
| Inflation Forecast Error | -0.048*** | -0.019 | -0.048*** | -0.018 | -0.048*** | -0.018 | -0.059*** | -0.044* | -0.060*** | -0.043* |
| | (0.010) | (0.013) | (0.010) | (0.014) | (0.010) | (0.014) | (0.018) | (0.025) | (0.018) | (0.025) |
| Trade Openness Forecast Error | 0.006 | 0.008 | 0.006 | 0.006 | 0.006 | 0.007 | 0.013** | 0.019*** | 0.013** | 0.019*** |
| | (0.005) | (0.005) | (0.004) | (0.005) | (0.005) | (0.005) | (0.007) | (0.007) | (0.007) | (0.007) |
| Operational Strength Index | -2.338*** | -2.395*** | -2.523*** | -2.360*** | -2.271*** | -2.406*** | -0.649 | -1.465** | -0.651 | -1.468** |
| | (0.573) | (0.758) | (0.618) | (0.791) | (0.583) | (0.761) | (0.460) | (0.658) | (0.461) | (0.659) |
| Constant | 1.363*** | 1.599*** | 2.427*** | 2.656** | 1.321*** | 1.597*** | 0.145 | 0.638 | 0.142 | 0.640* |
| | (0.330) | (0.434) | (0.660) | (1.239) | (0.336) | (0.435) | (0.269) | (0.386) | (0.269) | (0.383) |
| Observations | 258 | 229 | 258 | 229 | 258 | 229 | 751 | 665 | 751 | 665 |
| R2 (within for panel) | 0.356 | 0.201 | 0.367 | 0.217 | 0.365 | 0.204 | 0.224 | 0.180 | 0.225 | 0.186 |
| WEO vintage FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Tax-to-GDP Forecast Errors Over-Optimism | 0.382 | 0.559 | 0.382 | 0.559 | 0.382 | 0.559 | -0.123 | -0.081 | -0.123 | -0.081 |
| Over-Optimism (p-value) | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.038 | 0.256 | 0.038 | 0.256 |
| Effect of TP Yield Conditional on Revenue SCs (at Mean) | | | 0.313 | 0.413 | | | | | | |
| Conditional Effect (at Mean, p-value) | | | 0.032 | 0.008 | | | | | | |
| Effect of TP Yield Conditional on Revenue SCs (at Full) | | | 0.087 | -0.188 | | | | | | |
| Conditional Effect (at Full, p-value) | | | 0.811 | 0.533 | | | | | | |
| *Shapley Relative Explanatory Power* | | | | | | | | | | |
| *Over-Optimism in Yield and Tax Buoyancy* | 60.763 | 51.655 | 58.617 | 50.233 | 56.859 | 45.099 | 53.716 | 37.668 | 52.298 | 35.484 |
| TP Yield | 6.970 | 13.807 | 6.503 | 14.853 | | | 1.944 | 4.335 | | |
| Fully Implemented TP Yield | | | | | 3.574 | 8.528 | | | 1.574 | 3.284 |
| Tax Buoyancy Forecast Error | 53.793 | 37.848 | 52.113 | 35.380 | 53.285 | 36.571 | 51.773 | 33.332 | 50.725 | 32.200 |
| *Implementation Shortfall and Quality* | | | 1.926 | 4.755 | 4.603 | 7.692 | | | 1.809 | 5.008 |
| Implementation and Depth of Revenue SCs | | | 1.045 | 2.101 | | | | | | |
| Yield X Implementation and Depth of Revenue SCs | | | 0.881 | 2.653 | | | | | | |
| Implemented with Mod TP Yield | | | | | 0.990 | 5.521 | | | 0.137 | 0.618 |
| Not Implemented TP Yield | | | | | 3.613 | 2.171 | | | 1.672 | 4.390 |
| *Macro Errors* | 20.978 | 18.038 | 20.189 | 16.331 | 20.496 | 17.295 | 28.024 | 36.966 | 27.737 | 35.393 |
| Growth Forecast Error | 7.583 | 8.687 | 7.285 | 8.201 | 7.276 | 8.355 | 8.663 | 5.974 | 8.367 | 5.780 |
| Inflation Forecast Error | 9.609 | 1.189 | 9.316 | 1.005 | 9.385 | 1.125 | 3.149 | 2.861 | 3.199 | 2.607 |
| Trade Openness Forecast Error | 3.787 | 8.161 | 3.588 | 7.126 | 3.835 | 7.815 | 16.212 | 28.131 | 16.171 | 27.006 |
| *Other Factors* | 18.259 | 30.307 | 19.269 | 28.680 | 18.042 | 29.914 | 18.260 | 25.367 | 18.155 | 24.116 |
| Operational Strength Index | 9.501 | 15.761 | 9.808 | 14.126 | 9.138 | 15.741 | 4.462 | 12.369 | 4.458 | 12.094 |
| Cons + WEO vintage FE | 8.758 | 14.546 | 9.461 | 14.554 | 8.903 | 14.174 | 13.798 | 12.997 | 13.697 | 12.021 |

Note: OLS Estimates. Dependent variable is one- (h=1) or two-year-ahead (h=2) tax revenue-to-GDP forecast error for IMF-supported Programs (Columns (1)–(6)) and Surveillance (Columns (6)–(10)). In columns (3) and (4), tax policy yields and the index measuring the implementation and depth of revenue-related structural reforms are interacted. Columns (5), (6), (9) and (10) further distinguish tax policy measures by implementation status. Robust standard errors clustered by country are shown in parentheses. The use of *** denotes a coefficient significantly different from zero at the 0.01 confidence level; ** 0.05; * 0.10.

**Table 3. Channels: Aspects of Program Design, Combined Horizons h= 1, 2, 3**

| | (1) Baseline | (2) Tax | (3) Deficit | (4) Debt | (5) Growth |
|---|---|---|---|---|---|
| Tax Policy Yield | 1.194*** | 1.029*** | 1.170*** | 1.194*** | 1.191*** |
| | (0.262) | (0.205) | (0.283) | (0.263) | (0.238) |
| Implementation and Depth of Revenue SCs | -2.997 | -3.121 | -2.966 | -3.112 | -2.882 |
| | (2.835) | (2.981) | (2.822) | (2.846) | (2.819) |
| TP Yield X Implementation and Depth of Revenue SCs | -1.525*** | -1.471*** | -1.513*** | -1.528*** | -1.545*** |
| | (0.520) | (0.479) | (0.531) | (0.522) | (0.485) |
| Operational Strength Index | -1.933 | -1.028 | -1.984 | -2.112 | -1.913 |
| | (1.221) | (1.226) | (1.207) | (1.311) | (1.190) |
| Growth Forecast Error | 0.050* | 0.048* | 0.050* | 0.053* | 0.047* |
| | (0.027) | (0.027) | (0.027) | (0.028) | (0.028) |
| Tax Buoyancy Forecast Error | 0.567*** | 0.528*** | 0.564*** | 0.563*** | 0.561*** |
| | (0.113) | (0.113) | (0.111) | (0.112) | (0.112) |
| Inflation Forecast Error | -0.025 | -0.009 | -0.026 | -0.025 | -0.023 |
| | (0.034) | (0.032) | (0.034) | (0.034) | (0.035) |
| Trade Openness Forecast Error | 0.030*** | 0.024*** | 0.030*** | 0.030*** | 0.029*** |
| | (0.008) | (0.008) | (0.008) | (0.008) | (0.008) |
| Forecasted Chg. in Tax-to-GDP (t-1 to t+3) | | 0.315*** | | | |
| | | (0.112) | | | |
| Forecasted Chg. in Prim. Fiscal Balance (t-1 to t+3) | | | 0.015 | | |
| | | | (0.053) | | |
| Forecasted Chg. in Public Debt (t-1 to t+3) | | | | 0.006 | |
| | | | | (0.010) | |
| Forecasted Avg. Real GDP Growth (over t to t+3) | | | | | 0.080 |
| | | | | | (0.069) |
| Constant | 3.063* | 2.345 | 3.039* | 3.242* | 2.572 |
| | (1.701) | (1.910) | (1.691) | (1.743) | (1.762) |
| Observations | 246 | 246 | 246 | 246 | 246 |
| Number of countries | 51 | 51 | 51 | 51 | 51 |
| Number of programs | 84 | 84 | 84 | 84 | 84 |
| R-squared | 0.396 | 0.446 | 0.396 | 0.397 | 0.402 |
| WEO vintage FE | Yes | Yes | Yes | Yes | Yes |
| Tax-to-GDP Forecast Errors Over-Optimism | 0.597 | 0.597 | 0.597 | 0.597 | 0.597 |
| Over-Optimism (p-value) | 0.008 | 0.008 | 0.008 | 0.008 | 0.008 |

Note: OLS Estimates. Dependent variable is tax revenue-to-GDP forecast errors for the combined horizons covering the first three years following program approval (h=1, 2, 3). In all specifications, tax policy yields and the index measuring the implementation and depth of revenue-related structural reforms are interacted. Columns (2) to (6) add additional variables capturing aspects of program design. Robust standard errors clustered by country are shown in parentheses. The use of *** denotes a coefficient significantly different from zero at the 0.01 confidence level; ** 0.05; * 0.10.

## 4. Additional Robustness Checks and Sensitivity

We conduct several other robustness checks, focusing primarily on tax revenue-to-GDP forecast errors at program approval and specifications where tax policy interacts with the index of implementation and depth of structural reforms. The results are broadly unchanged. They are also confirmed in the extended panel of one- and two-year-ahead forecast errors, which includes IMF-supported program reviews as well as surveillance forecasts.

Appendix Table A.II.1 reports a set of robustness checks for the baseline results, including the exclusion of program with large GDP revisions,[20] of influential observations (based on Cook's distance and high-leverage

[20] To mitigate the influence of large statistical revisions, we exclude programs where GDP revisions between program approval and the latest WEO vintage used for outturns exceed 20 percent. This results in the exclusion of 17 Sub-Saharan African countries and 2 countries outside Africa. The main findings are robust to this restriction.

observations), robust regression estimates, and quantile regressions. Our baseline findings remain broadly unchanged, confirming that the results are not driven by large GDP revisions, influential observations or outliers. The quantile regressions provide additional insight into how these relationships vary across the distribution of forecast errors. Over-optimistic yield estimates and tax buoyancy assumptions contribute to forecast errors throughout the distribution. Higher implementation and depth of structural reforms are effective in reducing the contribution of optimistic tax policy yields to forecast errors in the lower and median quantiles. Stronger tax administration capacity plays an important role in reducing the largest forecast errors. Regarding macroeconomic factors, growth forecast errors are more relevant in the upper part of the distribution, while trade openness forecast errors remain positive and statistically significant across quantiles.

Appendix Table A.II.2 examines whether initial macroeconomic and fiscal conditions influence the size of tax revenue-to-GDP forecast errors under IMF-supported programs. The results show that the business cycle before and during the program, as well as the pre-program fiscal position—captured by tax revenue-to-GDP, debt-to-GDP, and fiscal balance-to-GDP—are not statistically associated with forecast errors. This suggests that the magnitude of tax revenue forecast errors is not primarily driven by the cyclical or fiscal conditions under which programs are initiated. Importantly, the baseline results remain robust.

Appendix Table A.II.3 incorporates variables capturing fiscal institutions and broader governance quality. The results show that the baseline findings remain largely unchanged. Among fiscal institutions, expenditure rules and budget balance rules are associated with smaller tax revenue-to-GDP forecast errors, suggesting that stronger fiscal frameworks may help improve the realism of revenue projections.[21] Similarly, when controlling for institutional quality indicators, regulatory quality is negatively and significantly associated with forecast errors.[22] Overall, these results confirm that while some institutional features may modestly influence forecast accuracy, the core determinants of tax revenue-to-GDP forecast errors discussed in our baseline results remain robust across specifications.

Appendix Table A.II.4 and Figure 3 show that the main findings remain broadly consistent across different country groups and program types. First, the size of tax revenue-to-GDP forecast errors is larger in LIDCs and PRGT programs. Second, across most subsamples, tax policy yield estimates remain positively associated with tax revenue-to-GDP forecast errors, while the interaction with the implementation and depth of revenue-related structural reforms remains negative, except for GRA programs where the average of forecast errors is not statistically different from zero. Similarly, tax buoyancy forecast errors remain a strong and robust determinant of tax revenue-to-GDP forecast errors. Stronger tax administration capacity reduces forecast errors in some specifications, while macroeconomic forecast errors—especially those related to growth and trade openness—continue to play a role in several subsamples. Finally, unexplained factors driving tax revenue-to-GDP forecast errors are more pronounced in LIDCs and PRGT programs. The Shapley decomposition associated with Appendix Table A.II.4 is also shown in Figure 3 (bottom panel of charts). It indicates that over-optimism in tax policy yield estimates and tax buoyancy assumptions explains a large share of the variance in revenue forecast errors across all groups. Macroeconomic forecast errors also account for a substantial share, except in GRA-supported programs, where unexplained factors and shocks play a larger role; these factors are also important for EMs.

[21] By contrast, revenue rules, debt rules, and fiscal councils are not statistically significant.

[22] Other governance indicators, including control of corruption, government effectiveness, rule of law, and conflict do not show significant effects

## B. Revenue-Gain Approach

Table 4 presents the results from the revenue-gain approach, where the change in the tax revenue-to-GDP ratio is regressed on the tax policy yield variable and its variants, controlling for macroeconomic and structural factors, and distinguishing between program and surveillance periods, consistent with the estimation of Equation (4).[23] Column (1) reports the materialization of revenue gains under IMF-supported programs. On average, a one percentage point of GDP increase in *ex-ante* expected tax policy yield translates into a 0.42 percentage point *ex-post* increase in the tax revenue-to-GDP ratio under IMF-supported programs. This coefficient is well below unity, indicating that a substantial portion—approximately 60 percent—of the projected revenue gains does not materialize in practice, be it due to implementation shortfalls, over-optimistic estimates of the yields from tax policy measures, or insufficient support by revenue-raising structural reforms. Column (2) allows for a comparison of the materialization of tax policy gains under surveillance. The coefficient on tax policy yield remains positive and statistically significant but is notably lower than under IMF-supported programs. The difference between the two coefficients is statistically significant. This suggests that, while IMF-supported programs embed greater revenue ambition as part of an intended fiscal regime shift—the expected revenue yields are 1.6 times higher than in surveillance—this ambition also delivers higher realized tax policy gains than surveillance cases.

Columns (3) and (4) restrict the tax policy yield variable to measures that were actually implemented, thereby abstracting from announced but unexecuted measures.[24] Relative to Columns (1) and (2), the estimated coefficients increase markedly, indicating a higher degree of forecast accuracy once implementation is accounted for. Under IMF-supported programs, a one percentage point of GDP increase in implemented tax policy yield is associated with a 0.53 percentage point increase in the tax revenue-to-GDP ratio, while the corresponding effect is 0.57 percentage point under surveillance (the difference is no longer statistically significant).[25] In both cases, the coefficients imply that more than half of the projected revenue gains from implemented measures do materialize, reducing implied over-optimism to below 50 percent. Taken together, these results underscore the important role of implementation in explaining gaps between projected and materialized tax revenues.

Column (5) introduces additional dimensions of implementation quality and depth based on the index of revenue-related structural reforms discussed above. The results reveal that both the depth and the effective implementation of revenue-related structural conditions are critical for maximizing revenue gains. In particular, the variable reflecting interacted implementation and depth of revenue-related structural conditions is strongly positive and significant, and the conditional materialization of tax policy yields under full implementation and high depth rises to 0.95 (see the line *"Effect of TP Yield Conditional on Revenue SCs (at Full)"*). This suggests that when revenue-related structural reforms are at high depth and fully implemented, the materialization of projected revenue gains from tax policy measures is nearly full. In contrast, when the reform's depth is average and reform implementation is at the sample mean, the materialization of tax policy yields (0.55) is not too different from the unconditional materialization of the yields from implemented measures in column (3). This

[23] Overall, this analysis confirms the findings discussed above using the forecast-error approach.

[24] Here "implemented" refers to both fully implemented and partially implemented measures.

[25] While the difference in the coefficient associated with the materialization rate between program and surveillance countries is no longer statistically significant, the difference in average ambition between the two cases implies that materialization is higher under programs, controlling for implementation.

finding highlights the central role of implementation and reform quality in bridging the gap between projected and actual tax revenue outcomes under IMF-supported programs.

We formally test whether each coefficient is statistically different from one (full materialization). In columns (1)–(4), the null hypothesis is decisively rejected (p=0.000), confirming that the shortfalls are not sampling artifacts and persist even when conditioning on implemented measures. In contrast, the conditional effect of 0.95 in column (5) is not statistically distinguishable from one: when revenue-related structural reforms are both fully implemented and of high depth, full materialization of projected yields cannot be rejected.

Column (6) further disaggregates the impact of tax policy measures by distinguishing between *large-measure* and *small-measure* revenue packages. The results indicate that only *large-measure* tax policy packages (where most of the yield comes from measures yielding more than 0.05 pp of GDP each) are associated with statistically significant revenue gains, while *small-measure* measures packages do not yield significant effects.[26] This finding underscores the limited effectiveness of fragmented or marginal tax policy changes and highlights the importance of ambitious, well-designed tax policy measures for achieving meaningful improvements in revenue mobilization.

Table 5 disaggregates the results of Table 4 by income groups (EMs vs. LIDCs) and by type of IMF lending facility (GRA vs. PRGT), revealing notable heterogeneity in both forecast accuracy and realized revenue gains. Across all specifications, the coefficient of the tax policy yield remains positive and statistically significant, but its magnitude varies substantially across groups (Panel A). EMs exhibit a stronger materialization of projected tax policy yields than LIDCs, with coefficients of 0.58 and 0.32, respectively, implying ex-ante over-optimism of roughly 40–70 percent. A similar pattern emerges when distinguishing between GRA and PRGT programs: GRA programs display a higher degree of revenue materialization (0.63) and lower implied over-optimism (37 percent), whereas PRGT programs show more muted effects (0.29) and substantially larger forecast gaps (over-optimism of 70 percent).[27] The formal tests of full materialization ($H_0$: coefficient = 1) are rejected across all subgroups in Panel A (p=0.000 throughout), confirming that over-optimism is statistically significant in every group of countries. When restricting attention to implemented measures (Panel B), the coefficients rise across all groups, underscoring the central role of execution in driving revenue outcomes. Implementation reduces over-optimism markedly—most notably under GRA programs, where the coefficient increases to 0.84 and over-optimism falls to 17 percent—while PRGT programs continue to exhibit weaker revenue materialization even after conditioning on implementation. The formal tests of full materialization ($H_0$: coefficient = 1) are rejected for most subgroups as well. With full implementation of higher depth revenue-related structural reforms, however, projected yields almost fully materialize (Panel C, line “Effect of TP Yield Conditional on Revenue SCs (at Full”), underscoring the central role of implementation and depth of measures to achieve the revenue targets. Overall, the table highlights that both income level and program type—as key determinants of the prevailing environment—shape the extent to which tax policy measures translate into actual revenue gains,

[26] The 0.05 percent of GDP threshold isolates measures too small to register at one decimal place. As a robustness check, we re-estimate using thresholds of 0.10 percent and the in-sample median. The large-measure coefficient remains positive, significant, and below one across all three thresholds (0.36, 0.55, and 0.60), confirming robust under-materialization. For small measures, the coefficient is not statistically different from zero at the baseline threshold but becomes significant at higher ones, suggesting that measures in the 0.05–0.10 percent range may yield some revenue—though materially less than larger reforms.

[27] In Panel A, over-optimism refers to both the expected implementation of agreed tax policy measures and the expected yields from these measures. In contrast, in Panel B over-optimism refers only to the expected yields of implemented measures. Comparing the two panels, one can infer that in GRA programs, over-optimism is broadly equally divided between expected policy implementation and the attendant yield estimates, while in PRGT programs what prevails is over-optimism regarding expected yields as implementation per se does not move the needle much.

with EMs and GRA programs consistently achieving higher materialization rates than LIDCs and PRGT programs. That said, consistent implementation of high-quality revenue-related structural measures sharply raises the materialized yields in both GRA and PRGT programs.

**Table 4. Tax Policy Yield and Actual Tax Revenue-to-GDP, IMF-Supported Programs and Surveillance**

| | (1) Program | (2) Surveillance | (3) Program | (4) Surveillance | (5) Program | (6) Program |
|---|---|---|---|---|---|---|
| Tax Policy Yield | 0.420*** | 0.273*** | | | | |
| | (0.030) | (0.030) | | | | |
| Tax Policy Yield Implemented | | | 0.528*** | 0.570*** | 0.283* | |
| | | | (0.086) | (0.088) | (0.150) | |
| Implementation and Depth of SCs | | | | | 1.188** | |
| | | | | | (0.480) | |
| TP Yield Implemented * Imp and Depth of SCs | | | | | 0.663*** | |
| | | | | | (0.195) | |
| Large TP Yield Implemented | | | | | | 0.356*** |
| | | | | | | (0.059) |
| Small TP Yield Implemented | | | | | | -0.600 |
| | | | | | | (3.296) |
| Real GDP growth | 0.011 | -0.024*** | 0.016 | -0.024** | 0.016 | 0.019 |
| | (0.012) | (0.006) | (0.013) | (0.008) | (0.014) | (0.014) |
| Inflation | -0.019 | -0.055*** | -0.017 | -0.053*** | -0.021 | -0.020 |
| | (0.014) | (0.009) | (0.014) | (0.009) | (0.014) | (0.013) |
| Terms-of-Trade growth | 0.005 | 0.002 | 0.003 | 0.002 | 0.005 | 0.004 |
| | (0.004) | (0.005) | (0.004) | (0.005) | (0.004) | (0.004) |
| Change in Trade openness | 3.516*** | 1.383 | 3.614*** | 1.310 | 3.613*** | 3.555*** |
| | (0.630) | (1.251) | (0.687) | (1.255) | (0.641) | (0.648) |
| Change in External debt | -1.829* | -0.596** | -1.676+ | -0.425* | -1.681+ | -1.769* |
| | (0.878) | (0.200) | (0.935) | (0.203) | (0.936) | (0.962) |
| Change in Agri | -0.018 | -0.047 | -0.004 | -0.048 | 0.007 | -0.020 |
| | (0.022) | (0.045) | (0.021) | (0.044) | (0.022) | (0.021) |
| Log (GDP per capita) | 0.236 | 0.409*** | -0.007 | 0.393*** | 0.169 | 0.103 |
| | (0.283) | (0.090) | (0.274) | (0.090) | (0.329) | (0.273) |
| Sq(Log (GDP per capita)) | -0.007 | -0.040** | 0.014 | -0.040** | -0.001 | 0.001 |
| | (0.026) | (0.013) | (0.026) | (0.014) | (0.033) | (0.026) |
| Within R-squared | 0.206 | 0.134 | 0.166 | 0.119 | 0.187 | 0.174 |
| Observations | 415 | 866 | 415 | 866 | 415 | 415 |
| #countries | 67 | 123 | 67 | 123 | 67 | 67 |
| p-value: H0 yield=1 (full materialization) | 0.000 | 0.000 | 0.000 | 0.0001 | – | |
| Effect of TP Yield Conditional on Revenue SCs (at Mean) | | | | | 0.547*** | |
| p-value: Cond. Effect at Mean | | | | | 0.000 | |
| Effect of TP Yield Conditional on Revenue SCs (at Full) | | | | | 0.946*** | |
| p-value: Cond. Effect at Full | | | | | 0.000 | |

Notes: Estimates are based on panel fixed-effects regressions on the annual change in tax revenue-to-GDP ratio. Country and year fixed effects are included, as well as a constant. Driscoll-Kraay standard errors. Growth variables are winsorized at the 5th and 95th percentiles. Changes in ratios are computed as t-(t-1); changes in levels are computed as t/(t-1) - t/(t-1)-1. + p < 0.15, * p < 0.10, ** p < 0.05, *** p < 0.010.

**Table 5. Robustness Checks by Country Income Level and by IMF Lending Facility**

| | (1) Programs | (2) EMs | (3) LIDCs | (4) GRA | (5) PRGT |
|---|---|---|---|---|---|
| Panel A | | | | | |
| Tax Policy Yield | 0.420*** | 0.576*** | 0.320*** | 0.630*** | 0.291** |
| | (0.030) | (0.086) | (0.086) | (0.068) | (0.104) |
| Within R-squared | 0.206 | 0.300 | 0.232 | 0.348 | 0.234 |
| p-value: H0 yield=1 (full materialization) | 0.000 | 0.001 | 0.000 | 0.000 | 0.000 |
| Panel B | | | | | |
| Tax Policy Yield Implemented | 0.528*** | 0.625** | 0.422+ | 0.835*** | 0.369 |
| | (0.086) | (0.219) | (0.244) | (0.204) | (0.260) |
| Within R-squared | 0.166 | 0.230 | 0.207 | 0.283 | 0.211 |
| p-value: H0 yield implemented=1 (full materialization) | 0.000 | 0.121 | 0.042 | 0.440 | 0.037 |
| Panel C | | | | | |
| Tax Policy Yield Implemented | 0.283* | 0.346 | 0.171 | 0.511 | 0.179 |
| | (0.150) | (0.281) | (0.292) | (0.359) | (0.292) |
| Impl and Depth of SCs | 1.188** | 3.848** | -0.259 | 2.223 | 0.317 |
| | (0.480) | (1.691) | (0.934) | (1.847) | (0.880) |
| TP Yield Implemented * Impl. and Depth of SCs | 0.663*** | 0.597 | 0.769* | 0.589 | 0.638 |
| | (0.195) | (0.381) | (0.358) | (0.457) | (0.416) |
| Within R-squared | 0.166 | 0.230 | 0.207 | 0.283 | 0.226 |
| Effect of TP Yield Conditional on Revenue SCs (at Mean) | 0.457*** | 0.473* | 0.408+ | 0.677** | 0.384 |
| Effect of TP Yield Conditional on Revenue SCs (at Full) | 0.946*** | 0.943** | 0.941* | 1.101** | 0.817* |
| Observations | 415 | 186 | 225 | 128 | 205 |
| Number of countries | 67 | 30 | 35 | 23 | 36 |

Notes: Estimates are based on panel fixed-effects regressions on the annual change in tax revenue-to-GDP ratio. Country and year fixed effects are included, as well as a constant. Driscoll-Kraay standard errors. Growth variables are winsorized at the 5th and 95th percentiles. Changes in ratios are computed as t-(t-1); changes in levels are computed as t/(t-1) - t/(t-1)-1. + p < 0.15, * p < 0.10, ** p < 0.05, *** p < 0.010. GRA-PRGT classifications are based on the resource account from which Fund-supported programs draw, while the EM–LIDC classification follows the IMF World Economic Outlook (WEO). The two classifications overlap but are not identical: some PRGT eligible countries are classified as emerging markets (EMs) under the WEO classification, and some countries classified as LIDCs in the WEO are not PRGT eligible. Direct links between the LIDC grouping and PRGT supported programs should therefore be avoided.

# V. Conclusions

IMF-supported programs in 2015-24 forecasted a median increase in the tax revenue-to-GDP ratio of 1 percentage point by the third program year, but the median actual increase was only 0.3 percentage point. To explain this large gap, this paper examines the drivers of tax revenue-to-GDP forecast errors under IMF-supported programs, with a particular focus on the role of tax policy yield estimates, tax buoyancy assumptions, institutional capacity, and macroeconomic forecast errors. Leveraging novel datasets on tax policy yields (TAPRED) and tax administration capacity, we document a clear pattern of over-optimism in tax revenue-to-GDP forecasts under IMF-supported programs.

Our results show that tax revenue-to-GDP forecasts are consistently more optimistic when compared to actual outcomes. A key finding is that over-optimism in tax policy yield estimation, policy implementation expectations, and tax buoyancy assumptions plays a central role in explaining these forecast errors, alongside macroeconomic forecast errors and a residual optimism. By contrast, stronger tax administration capacity reduces forecast errors, highlighting the importance of institutional capacity in translating tax policy measures into realized revenue gains and improving realism. More broadly, the results suggest that a large share of projected tax policy yields—often more than half—fails to materialize in practice due to variety of reasons, but the higher ambition in these yields in IMF-supported programs is also associated with higher realized tax policy

gains than in surveillance cases. At the same time, our findings show that this gap between forecasts and outcomes is not inevitable. When revenue measures are well-designed, fully implemented, and accompanied by high-depth revenue-related structural reforms, projected revenue gains materialize to a large extent. This implies that over-optimism emerges primarily when ambitious revenue targets are not supported by sufficiently deep and well-implemented reforms.

Three clear avenues for future research emerge. First, political conditions—such as electoral cycles and political instability—may shape the credibility and implementation of revenue measures, thereby amplifying forecast errors. Second, disaggregating tax revenue-to-GDP into its main components and mapping each to its underlying base (imports, consumption, wages, profits) would help assess whether forecast errors vary systematically across tax categories and better identify their drivers. Third, tax evasion and compliance behavior may play an important role in explaining the shortfalls in tax policy yields. These are promising directions for future work, especially as more granular data become available.

# Appendix I. Data Description, and Summary Statistics

**Table A.I.1. Data Description and Sources**

| Variables | Descriptions | Sources |
|---|---|---|
| Tax policy yield | Narrative measure of tax policy yield using LLM (TAPRED) as percentage of GDP. | Atsebi and Goyal (2026) |
| Tax revenues (% of GDP) | Total tax revenues as a percentage of GDP. | Authors' calculations based on the IMF's WEO |
| Tax buoyancy | Change in nominal tax revenue as a share of change in nominal GDP | Authors' calculations based on IMF WEO |
| Quality index of revenue-related structural reforms | The depth and implementation of structural benchmarks under IMF-supported program | MONA |
| Operational Strength Index (OSI) of tax administration | An indicator measuring the overall strength of tax administration | Atsebi et al. (2025) |
| Primary fiscal balance | Primary balance as percentage of GDP | Authors' calculations based on IMF WEO |
| Public debt (% GDP) | Public debt as percentage of GDP | Authors' calculations based on IMF WEO |
| Fiscal rules | Description of fiscal rules | IMF Fiscal Rule database |
| Governance indicators | Indicators measuring the level of governance | World Bank Governance Indicators (WGI) |
| Real GDP growth | Real GDP growth | Authors' calculations based on IMF WEO |
| Log GDP per capita | Log GDP per capita (USD) | Authors' calculations based on IMF WEO |
| Squared log GDP per capita | Squared log GDP per capita (USD) | Authors' calculations based on IMF WEO |
| Trade openness (% GDP) | Trade openness (exports + imports/GDP) | Authors' calculations based on IMF WEO |
| External debt (% GDP) | External debt as percentage of GDP | Authors' calculations based on IMF WEO and WB WDI |
| CPI inflation | CPI inflation | Authors' calculations based on IMF WEO and WB WDI |
| Terms of trade (2000=1, lagged) | Terms of trade | Authors' calculations based on IMF WEO and WB WDI |
| Oil exports (% GDP) | Oil exports as percentage of GDP | Authors' calculations based on IMF WEO |
| Agriculture share of GDP | Agriculture value added share of GDP | Authors' calculations based on WB WDI |

**Table A.I.2. Summary Statistics of Variables Used in Equations (2) and (3)**

| | Obs. | Mean | Std Dev. | Min | Max |
|---|---|---|---|---|---|
| **(A) At Program Approval** | | | | | |
| Tax-to-GDP Forecast Error (h=1) | 92 | 0.376 | 1.448 | -3.44 | 5.56 |
| Tax-to-GDP Forecast Error (h=1, 2, 3) | 282 | 0.628 | 1.778 | -3.440 | 5.758 |
| Tax Policy Yield | 291 | 0.297 | 0.539 | -1.400 | 2.467 |
| Fully Implemented TP Yield | 291 | 0.167 | 0.507 | -3.000 | 1.940 |
| Implemented with Mod TP Yield | 291 | 0.067 | 0.231 | -0.760 | 1.600 |
| Not Implemented TP Yield | 291 | 0.054 | 0.205 | -0.400 | 2.100 |
| Implementation and Depth of Revenue SCs | 294 | 0.415 | 0.270 | 0.000 | 0.750 |
| Tax Buoyancy Forecast Error | 279 | 0.230 | 1.020 | -1.732 | 2.186 |
| Growth Forecast Error | 294 | 1.045 | 3.908 | -8.752 | 13.022 |
| Inflation Forecast Error | 291 | -1.903 | 4.565 | -14.568 | 16.907 |
| Trade Openness Forecast Error | 288 | 3.673 | 15.412 | -38.516 | 71.860 |
| Operational Strength Index | 264 | 0.527 | 0.138 | 0.208 | 0.745 |
| Avg. Pre Output Gap to GDP [h-2, h-1] | 294 | -1.310 | 2.386 | -10.138 | 3.863 |
| Avg. Post Output Gap to GDP [h, h+3] | 294 | -1.235 | 1.164 | -4.771 | 0.573 |
| Avg. Pre Tax-to-GDP [h-2, h-1] | 282 | 16.316 | 6.120 | 1.452 | 29.423 |
| Avg. Pre Debt to GDP [h-2, h-1] | 288 | 58.391 | 25.569 | 15.081 | 177.989 |
| Avg. Pre Fiscal Balance to GDP [h-2, h-1] | 288 | -2.219 | 2.849 | -9.369 | 6.627 |
| ER Fiscal Rule | 294 | 0.255 | 0.437 | 0.000 | 1.000 |
| BBR Fiscal Rule | 294 | 0.605 | 0.490 | 0.000 | 1.000 |
| Regulatory Quality | 291 | -0.357 | 0.624 | -1.900 | 1.032 |
| **(B) Panel of One- and Two-year-ahead Forecast Errors (Include Reviews and Surveillance)** | | | | | |
| Tax-to-GDP Forecast Error (h=1) | 1468 | -0.017 | 1.733 | -5.306 | 5.957 |
| Tax-to-GDP Forecast Error (h=2) | 1297 | 0.086 | 1.962 | -5.306 | 5.957 |
| Tax Policy Yield | 1305 | 0.083 | 0.482 | -1.383 | 2.000 |
| Implemented Measures Annual Yield | 1305 | 0.023 | 0.486 | -5.000 | 3.762 |
| Implemented Mod Measures Annual Yield | 1305 | 0.021 | 0.170 | -1.250 | 2.000 |
| Not Implemented Measures Annual Yield | 1305 | 0.035 | 0.220 | -1.000 | 3.150 |
| Implementation and Depth of Revenue SCs | 1501 | 0.079 | 0.204 | 0.000 | 1.000 |
| Tax Buoyancy Forecast Error | 1297 | 0.108 | 1.241 | -2.478 | 2.885 |
| Growth Forecast Error | 1470 | 1.054 | 4.454 | -10.158 | 13.626 |
| Inflation Forecast Error | 1463 | -2.039 | 7.256 | -38.966 | 14.675 |
| Trade Openness Forecast Error | 1376 | 1.353 | 19.782 | -77.833 | 84.348 |
| Operational Strength Index | 1133 | 0.540 | 0.142 | 0.023 | 0.859 |

Notes: Panel (A) reports summary statistics for the sample focusing on program approval. In this panel, explanatory variables correspond to the three years following program approval. Panel (B) reports summary statistics for the panel of one- and two-year-ahead forecast errors (including reviews and surveillance). In this panel, explanatory variables correspond to the two-year-ahead forecast horizon.

## Table A.I.3. Correlations of Variables Used in Equations (2) and (3)

### (A) At Program Approval

| Variable | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) | (13) | (14) | (15) | (16) | (17) | (18) | (19) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (1) Tax-to-GDP Forecast Error (h=1,2,3) | 1.00 | | | | | | | | | | | | | | | | | | |
| (2) Tax Policy Yield | 0.14* | 1.00 | | | | | | | | | | | | | | | | | |
| (3) Fully Implemented TP Yield | 0.09 | 0.82* | 1.00 | | | | | | | | | | | | | | | | |
| (4) Implemented with Mod TP Yield | 0.13* | 0.35* | -0.06 | 1.00 | | | | | | | | | | | | | | | |
| (5) Not Implemented TP Yield | 0.01 | 0.35* | -0.01 | -0.05 | 1.00 | | | | | | | | | | | | | | |
| (6) Implementation and Depth of Revenue SCs | 0.14* | 0.02 | -0.03 | 0.04 | 0.05 | 1.00 | | | | | | | | | | | | | |
| (7) Tax Buoyancy Forecast Error | 0.37* | -0.07 | -0.04 | 0.03 | -0.13* | 0.13* | 1.00 | | | | | | | | | | | | |
| (8) Growth Forecast Error | 0.21* | -0.12* | -0.17* | 0.03 | -0.01 | 0.02 | 0.07 | 1.00 | | | | | | | | | | | |
| (9) Inflation Forecast Error | -0.10 | -0.07 | -0.05 | -0.03 | -0.07 | -0.01 | -0.02 | -0.18* | 1.00 | | | | | | | | | | |
| (10) Trade Openness Forecast Error | 0.32* | -0.05 | -0.06 | 0.04 | -0.05 | 0.02 | 0.03 | 0.25* | 0.09 | 1.00 | | | | | | | | | |
| (11) Operational Strength Index | -0.24* | -0.11* | -0.04 | -0.04 | -0.15* | -0.41* | 0.02 | -0.04 | -0.24* | -0.19* | 1.00 | | | | | | | | |
| (12) Avg. Pre Output Gap to GDP [h-2,h-1] | -0.16* | -0.09 | -0.05 | -0.04 | -0.11* | 0.08 | 0.08 | -0.05 | 0.09 | -0.17* | 0.16* | 1.00 | | | | | | | |
| (13) Avg. Post Output Gap to GDP [h,h+3] | 0.09 | -0.05 | 0.05 | -0.10 | -0.09 | -0.13* | -0.03 | 0.07 | 0.10* | 0.12* | 0.15* | 0.08 | 1.00 | | | | | | |
| (14) Avg. Pre Tax-to-GDP [h-2,h-1] | -0.11* | -0.02 | 0.05 | -0.08 | -0.10 | -0.22* | -0.22* | 0.07 | 0.05 | -0.01 | 0.23* | 0.04 | 0.16* | 1.00 | | | | | |
| (15) Avg. Pre Debt to GDP [h-2,h-1] | 0.16* | 0.10* | 0.02 | -0.02 | 0.22* | 0.14* | -0.01 | 0.24* | -0.18* | 0.21* | -0.16* | -0.06 | -0.03 | 0.11* | 1.00 | | | | |
| (16) Avg. Pre Fiscal Balance to GDP [h-2,h-1] | -0.13* | -0.08 | -0.02 | -0.02 | -0.11* | -0.06 | -0.15* | -0.01 | 0.11* | -0.04 | -0.14* | 0.04 | 0.02 | 0.26* | 0.09 | 1.00 | | | |
| (17) ER Fiscal Rule | -0.32* | -0.16* | -0.11* | -0.10* | -0.10* | -0.27* | -0.07 | -0.08 | -0.07 | -0.18* | 0.48* | 0.15* | 0.06 | 0.18* | -0.14* | 0.15* | 1.00 | | |
| (18) BBR Fiscal Rule | -0.17* | -0.08 | -0.07 | 0.04 | -0.12* | -0.21* | -0.04 | -0.09 | 0.16* | -0.06 | 0.08 | 0.05 | 0.06 | -0.01 | -0.24* | 0.20* | 0.38* | 1.00 | |
| (19) Regulatory Quality | -0.26* | -0.12* | -0.06 | -0.07 | -0.12* | -0.36* | -0.05 | -0.07 | 0.15* | -0.20* | 0.56* | 0.15* | 0.20* | 0.61* | -0.10 | 0.07 | 0.38* | 0.24* | 1.00 |

### (B) Panel of One- and Two-year-ahead Forecast Errors (Include Reviews and Surveillance)

| Variable | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (1) Tax-to-GDP Forecast Error (h=2) | 1 | | | | | | | | | | |
| (2) Tax Policy Yield | 0.18* | 1 | | | | | | | | | |
| (3) Implemented Measures Annual Yield | 0.13* | 0.80* | 1 | | | | | | | | |
| (4) Implemented Mod Measures Annual Yield | 0.07* | 0.34* | -0.02 | 1 | | | | | | | |
| (5) Not Implemented Measures Annual Yield | 0.09* | 0.42* | 0 | 0.05* | 1 | | | | | | |
| (6) Implementation and Depth of Revenue SCs | 0.12* | 0.13* | 0.11* | 0.06* | -0.01 | 1 | | | | | |
| (7) Tax Buoyancy Forecast Error | 0.23* | 0.06* | 0.01 | 0.04 | 0.07* | 0.03 | 1 | | | | |
| (8) Growth Forecast Error | 0.16* | 0.06* | 0.04 | 0.04 | 0.04 | -0.01 | 0.03 | 1 | | | |
| (9) Inflation Forecast Error | -0.09* | -0.02 | 0.03 | -0.08* | -0.04 | 0 | -0.05* | -0.11* | 1 | | |
| (10) Trade Openness Forecast Error | 0.22* | 0.13* | 0.13* | 0.05 | -0.01 | 0.02 | 0 | 0.18* | 0.18* | 1 | |
| (11) Operational Strength Index | -0.14* | -0.20* | -0.17* | -0.06* | -0.07* | -0.16* | -0.03 | -0.08* | -0.12* | -0.13* | 1 |

Notes: Panel (A) reports correlations between the tax revenue-to-GDP forecast error (h=1,2,3) and explanatory variables for the sample focusing on program approval. In this panel, explanatory variables correspond to the three years following program approval. Panel (B) reports correlations between the tax revenue-to-GDP forecast error (h=2) and explanatory variables for the panel of one- and two-year-ahead forecast errors (including reviews and surveillance). In this panel, explanatory variables correspond to the two-year-ahead forecast horizon.

**Table A.I.4. Summary Statistics of Variables Used in Equation (4)**

| | Obs. | Mean | Std Dev. | Min | Max |
|---|---|---|---|---|---|
| Change in Tax Revenue-to-GDP | 1733 | 0.067 | 1.010 | -1.887 | 1.968 |
| Tax Policy Yield | 1700 | 0.141 | 0.708 | -7.589 | 8.100 |
| Tax Policy Yield Implemented | 1700 | 0.058 | 0.265 | -0.430 | 0.700 |
| Implementation and Depth of Revenue SCs | 1990 | 0.067 | 0.170 | 0.000 | 0.857 |
| TP Yield X Implementation. and Depth of Revenue SCs | 1700 | 0.027 | 0.126 | -0.664 | 1.591 |
| Large TP Yield Implemented | 1700 | 0.057 | 0.584 | -7.589 | 7.700 |
| Small TP Yield Implemented | 1700 | 0.002 | 0.013 | -0.095 | 0.163 |
| Real GDP growth | 1885 | 2.915 | 3.322 | -3.661 | 9.037 |
| Inflation | 1883 | 4.859 | 5.084 | -0.055 | 18.222 |
| Terms-of-trade growth | 1663 | 0.781 | 7.134 | -11.608 | 16.380 |
| Change in Trade openness | 1882 | -0.000 | 0.076 | -0.144 | 0.144 |
| Change in External debt | 1723 | -0.003 | 0.074 | -0.164 | 0.144 |
| Change in Agriculture | 1772 | -0.056 | 0.805 | -2.015 | 1.653 |
| Log (GDP per capita, USD) | 1885 | 8.325 | 2.546 | -22.470 | 16.192 |
| Sq(Log (GDP per capita, USD)) | 1885 | 75.790 | 35.839 | 0.010 | 504.879 |

**Table A.I.5. Correlations of Variables Used in Equation (4)**

| Variable | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (1) Change in Tax-to-GDP | 1 | | | | | | | | | | | |
| (2) Tax Policy Yield | 0.17* | 1 | | | | | | | | | | |
| (3) Tax Policy Yield Implemented | 0.14* | 0.68* | 1 | | | | | | | | | |
| (4) Large TP Yield Implemented | 0.16* | 0.85* | 0.78* | 1 | | | | | | | | |
| (5) Small TP Yield Implemented | 0.01 | 0.09* | 0.17* | 0.08* | 1 | | | | | | | |
| (6) Real GDP growth | 0.15* | 0.02 | 0.06* | 0.06* | 0.05* | 1 | | | | | | |
| (7) Inflation | 0.00 | 0.12* | 0.12* | 0.11 | 0.02 | 0.08* | 1 | | | | | |
| (8) Terms-of-trade growth | 0.07* | 0.06* | 0.10* | 0.06 | 0.01 | 0.08* | 0.04* | 1 | | | | |
| (9) Change in Trade openness | 0.18* | 0.00 | 0.00 | 0.02 | 0.00 | 0.22* | 0.08* | -0.01 | 1 | | | |
| (10) Change in external debt | -0.08* | 0.04* | 0.05* | 0.04 | 0.02 | -0.24* | -0.07* | -0.12* | -0.05* | 1 | | |
| (11) Change in agriculture | -0.13* | -0.04* | -0.03 | -0.03 | 0.00 | -0.21* | 0.00 | -0.08* | -0.11* | 0.07* | 1 | |
| (12) Log (GDP per capita, USD) | -0.04* | -0.13* | -0.13* | -0.09* | -0.09* | -0.12* | -0.43* | -0.04* | 0.01 | 0.01* | 0.05* | 1 |

# Appendix II. Additional Tables

**Table A.II.1. Robustness Checks: Outliers and Quantile Regressions, Combined Horizons h= 1, 2, 3**

| | (1) Baseline | (2) GDP Revision | (3) No Cook's | (4) No Leverage | (5) Rob. Reg | (6) Qreg: Q1 | (7) Qreg: Q2 | (8) Qreg: Q3 |
|---|---|---|---|---|---|---|---|---|
| Tax Policy Yield | 1.194*** | 1.235*** | 0.804** | 1.197*** | 1.238*** | 1.286*** | 1.366*** | 0.995** |
| | (0.262) | (0.378) | (0.319) | (0.261) | (0.372) | (0.189) | (0.447) | (0.489) |
| Implementation and Depth of Revenue SCs | -2.997 | 0.827 | -3.987** | -2.998 | -3.347 | -7.350*** | -4.968*** | 0.936 |
| | (2.829) | (4.296) | (1.836) | (2.820) | (2.429) | (1.231) | (1.693) | (3.909) |
| TP Yield X Implementation. and Depth of Revenue SCs | -1.525*** | -1.548** | -0.687 | -1.530*** | -1.535** | -1.603*** | -1.962** | -0.906 |
| | (0.519) | (0.716) | (0.599) | (0.517) | (0.705) | (0.364) | (0.797) | (0.893) |
| Operational Strength Index | -1.933 | -2.084 | -2.507*** | -1.959 | -2.378*** | 0.223 | -2.535*** | -3.615*** |
| | (1.219) | (1.490) | (0.893) | (1.217) | (0.861) | (0.723) | (0.899) | (0.775) |
| Growth Forecast Error | 0.050* | 0.041 | 0.031 | 0.050* | 0.034 | 0.023 | 0.065** | 0.071** |
| | (0.027) | (0.032) | (0.022) | (0.027) | (0.023) | (0.022) | (0.026) | (0.028) |
| Tax Buoyancy Forecast Error | 0.567*** | 0.550*** | 0.623*** | 0.574*** | 0.564*** | 0.629*** | 0.432*** | 0.558*** |
| | (0.113) | (0.140) | (0.101) | (0.113) | (0.089) | (0.085) | (0.105) | (0.090) |
| Inflation Forecast Error | -0.025 | -0.008 | -0.034 | -0.028 | -0.019 | 0.007 | -0.000 | -0.045 |
| | (0.034) | (0.036) | (0.024) | (0.035) | (0.024) | (0.027) | (0.024) | (0.038) |
| Trade Openness Forecast Error | 0.030*** | 0.023 | 0.038*** | 0.030*** | 0.037*** | 0.035*** | 0.032*** | 0.029*** |
| | (0.008) | (0.015) | (0.007) | (0.008) | (0.006) | (0.005) | (0.007) | (0.006) |
| Constant | 3.063* | 0.995 | 3.685*** | 3.083* | 4.465*** | 3.732*** | 4.232*** | 1.859 |
| | (1.698) | (2.530) | (1.170) | (1.689) | (1.681) | (0.890) | (0.986) | (2.173) |
| Observations | 246 | 195 | 225 | 240 | 246 | 246 | 246 | 246 |
| Number of countries | 51 | 40 | 50 | 51 | 51 | 51 | 51 | 51 |
| Number of programs | 84 | 67 | 81 | 82 | 84 | 84 | 84 | 84 |
| R-squared | 0.396 | 0.346 | 0.482 | 0.374 | 0.425 | 0.281 | 0.244 | 0.271 |
| WEO vintage FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Tax-to-GDP Forecast Errors Over-Optimism | 0.597 | 0.446 | 0.529 | 0.637 | 0.597 | -0.445 | 0.520 | 1.609 |
| Over-Optimism (p-value) | 0.008 | 0.069 | 0.015 | 0.006 | 0.000 | 0.004 | 0.000 | 0.000 |

Note: OLS Estimates (columns (1)–(4)), Robust Regression (column (5)), and Quantile Regressions (columns (6)–(8)). Dependent variable is tax revenue-to-GDP forecast errors for the combined horizons covering the first three years following program approval (h=1, 2, 3). In all specifications, tax policy yields and the index measuring the implementation and depth of revenue-related structural reforms are interacted. Robust standard errors clustered by country are shown in parentheses. The use of *** denotes a coefficient significantly different from zero at the 0.01 confidence level; ** 0.05; * 0.10.

**Table A.II.2. Robustness Checks: Initial Conditions and Post-Program Business Cycle, Combined Horizons h= 1, 2, 3**

| | (1) Baseline | (2) PreOgap | (3) PostOgap | (4) PreTax | (5) PreDebt | (6) PreDebt |
|---|---|---|---|---|---|---|
| Tax Policy Yield | 1.194*** | 1.234*** | 1.205*** | 1.261*** | 1.201*** | 1.136*** |
| | (0.262) | (0.242) | (0.267) | (0.309) | (0.260) | (0.263) |
| Implementation and Depth of Revenue SCs | -2.997 | -2.186 | -3.629 | -2.785 | -2.924 | -3.173 |
| | (2.835) | (2.716) | (2.985) | (2.806) | (2.827) | (2.906) |
| TP Yield X Implementation and Depth of Revenue SCs | -1.525*** | -1.544*** | -1.547*** | -1.656*** | -1.541*** | -1.473*** |
| | (0.520) | (0.491) | (0.537) | (0.579) | (0.522) | (0.525) |
| Operational Strength Index | -1.933 | -1.266 | -2.211 | -1.679 | -1.901 | -2.080* |
| | (1.221) | (1.293) | (1.358) | (1.280) | (1.229) | (1.166) |
| Growth Forecast Error | 0.050* | 0.052* | 0.044* | 0.054* | 0.050* | 0.052* |
| | (0.027) | (0.028) | (0.026) | (0.028) | (0.028) | (0.027) |
| Tax Buoyancy Forecast Error | 0.567*** | 0.577*** | 0.573*** | 0.540*** | 0.568*** | 0.556*** |
| | (0.113) | (0.116) | (0.114) | (0.111) | (0.113) | (0.112) |
| Inflation Forecast Error | -0.025 | -0.015 | -0.026 | -0.021 | -0.025 | -0.023 |
| | (0.034) | (0.036) | (0.035) | (0.034) | (0.034) | (0.035) |
| Trade Openness Forecast Error | 0.030*** | 0.028*** | 0.030*** | 0.030*** | 0.029*** | 0.029*** |
| | (0.008) | (0.009) | (0.009) | (0.008) | (0.008) | (0.008) |
| Avg. Pre Output Gap to GDP [h-2, h-1] | | -0.103* | | | | |
| | | (0.055) | | | | |
| Avg. Post Output Gap to GDP [h, h+3] | | | 0.166 | | | |
| | | | (0.162) | | | |
| Avg. Pre Tax-to-GDP [h-2, h-1] | | | | -0.023 | | |
| | | | | (0.022) | | |
| Avg. Pre Debt to GDP [h-2, h-1] | | | | | 0.002 | |
| | | | | | (0.005) | |
| Avg. Pre Fiscal Balance to GDP [h-2, h-1] | | | | | | -0.048 |
| | | | | | | (0.061) |
| Constant | 3.063* | 2.114 | 3.785* | 3.205* | 2.910* | 3.157* |
| | (1.701) | (1.703) | (2.030) | (1.638) | (1.687) | (1.741) |
| Observations | 246 | 246 | 246 | 246 | 246 | 246 |
| Number of countries | 51 | 51 | 51 | 51 | 51 | 51 |
| Number of programs | 84 | 84 | 84 | 84 | 84 | 84 |
| R-squared | 0.396 | 0.408 | 0.404 | 0.400 | 0.396 | 0.399 |
| WEO vintage FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Tax-to-GDP Forecast Errors Over-Optimism | 0.597 | 0.597 | 0.597 | 0.597 | 0.597 | 0.597 |
| Over-Optimism (p-value) | 0.008 | 0.008 | 0.008 | 0.008 | 0.008 | 0.008 |

Note: OLS Estimates. Dependent variable is tax revenue-to-GDP forecast errors for the combined horizons covering the first three years following program approval (h=1, 2, 3). In all specifications, tax policy yields and the index measuring the implementation and depth of revenue-related structural reforms are interacted. Columns (2) to (6) add additional variables capturing initial economic and fiscal conditions, as well as post-program business cycle. Robust standard errors clustered by country are shown in parentheses. The use of *** denotes a coefficient significantly different from zero at the 0.01 confidence level; ** 0.05; * 0.10.

**Table A.II.3. Robustness Checks: Fiscal Institutions and Governance Indicators, Combined Horizons h= 1, 2, 3**

| | (1) Baseline | (2) ER rule | (3) BBR rule | (4) Reg. Quality |
|---|---|---|---|---|
| Tax Policy Yield | 1.194*** | 0.983*** | 1.198*** | 1.218*** |
| | (0.262) | (0.220) | (0.256) | (0.278) |
| Implementation and Depth of Revenue SCs | -2.997 | -2.465 | -2.558 | -2.728 |
| | (2.835) | (3.259) | (3.416) | (2.819) |
| TP Yield X Implementation and Depth of Revenue SCs | -1.525*** | -1.147** | -1.543*** | -1.637*** |
| | (0.520) | (0.490) | (0.520) | (0.555) |
| Operational Strength Index | -1.933 | -0.512 | -1.706 | -0.599 |
| | (1.221) | (1.299) | (1.237) | (1.406) |
| Growth Forecast Error | 0.050* | 0.045* | 0.047* | 0.052* |
| | (0.027) | (0.026) | (0.026) | (0.028) |
| Tax Buoyancy Forecast Error | 0.567*** | 0.538*** | 0.568*** | 0.566*** |
| | (0.113) | (0.109) | (0.106) | (0.110) |
| Inflation Forecast Error | -0.025 | -0.023 | -0.012 | -0.009 |
| | (0.034) | (0.038) | (0.033) | (0.035) |
| Trade Openness Forecast Error | 0.030*** | 0.027*** | 0.028*** | 0.027*** |
| | (0.008) | (0.008) | (0.009) | (0.008) |
| ER Fiscal Rule | | -0.810** | | |
| | | (0.381) | | |
| BBR Fiscal Rule | | | -0.621** | |
| | | | (0.295) | |
| Regulatory Quality | | | | -0.572** |
| | | | | (0.217) |
| Constant | 3.063* | 2.239 | 3.091 | 2.073 |
| | (1.701) | (1.992) | (2.003) | (1.820) |
| Observations | 246 | 246 | 246 | 242 |
| Number of countries | 51 | 51 | 51 | 51 |
| Number of programs | 84 | 84 | 84 | 83 |
| R-squared | 0.396 | 0.423 | 0.416 | 0.395 |
| WEO vintage FE | Yes | Yes | Yes | Yes |
| Tax-to-GDP Forecast Errors Over-Optimism | 0.597 | 0.597 | 0.597 | 0.636 |
| Over-Optimism (p-value) | 0.008 | 0.008 | 0.008 | 0.006 |

Note: OLS Estimates. Dependent variable is tax revenue-to-GDP forecast errors for the combined horizons covering the first three years following program approval (h=1, 2, 3). In all specifications, tax policy yields and the index measuring the implementation and depth of revenue-related structural reforms are interacted. Columns (2) to (6) add additional variables capturing initial economic and fiscal conditions, as well as post-program business cycle. Robust standard errors clustered by country are shown in parentheses. The use of *** denotes a coefficient significantly different from zero at the 0.01 confidence level; ** 0.05; * 0.10.

## Table A.II.4. Robustness Checks: Different Samples, Combined Horizons h = 1, 2, 3

| | (1) Baseline | (2) EMs | (3) LIDCs | (4) Non FCS | (5) Non Fuel Exp | (6) GRA | (7) PRGT | (8) Non-Dirb | (9) Completed |
|---|---|---|---|---|---|---|---|---|---|
| Tax Policy Yield | 1.194*** | 1.079* | 0.763*** | 1.365*** | 1.352*** | 0.539 | 0.755*** | 1.388*** | 1.236*** |
| | (0.262) | (0.531) | (0.157) | (0.349) | (0.287) | (1.689) | (0.136) | (0.420) | (0.419) |
| Implementation and Depth of Revenue SCs | -2.997 | 1.461 | -3.735 | -4.797 | -3.749 | 5.070 | -4.003 | 3.575 | -1.464 |
| | (2.835) | (5.715) | (2.678) | (2.967) | (2.293) | (12.023) | (2.852) | (4.728) | (5.051) |
| TP Yield X Implementation and Depth of Revenue SCs | -1.525*** | -0.898 | -0.820** | -1.921*** | -1.674*** | 0.062 | -0.678* | -1.812* | -1.431** |
| | (0.520) | (1.052) | (0.395) | (0.682) | (0.582) | (3.081) | (0.389) | (0.947) | (0.672) |
| Operational Strength Index | -1.933 | -0.197 | -1.497 | -0.333 | -4.117*** | 2.641 | -1.358 | -7.295* | -3.964** |
| | (1.221) | (2.051) | (1.218) | (1.354) | (1.267) | (2.961) | (1.289) | (3.529) | (1.638) |
| Growth Forecast Error | 0.050* | 0.018 | 0.143** | 0.028 | 0.065** | -0.016 | 0.200*** | 0.023 | 0.030 |
| | (0.027) | (0.033) | (0.059) | (0.031) | (0.028) | (0.035) | (0.054) | (0.044) | (0.040) |
| Tax Buoyancy Forecast Error | 0.567*** | 0.438** | 0.485*** | 0.566*** | 0.566*** | 0.239 | 0.454*** | 0.502** | 0.716*** |
| | (0.113) | (0.195) | (0.106) | (0.172) | (0.117) | (0.138) | (0.103) | (0.211) | (0.120) |
| Inflation Forecast Error | -0.025 | 0.010 | -0.044 | -0.028 | -0.075** | 0.021 | -0.041 | 0.044 | 0.021 |
| | (0.034) | (0.049) | (0.047) | (0.035) | (0.030) | (0.042) | (0.051) | (0.034) | (0.056) |
| Trade Openness Forecast Error | 0.030*** | 0.047*** | 0.011 | 0.033*** | 0.033*** | 0.063** | 0.009 | 0.027 | 0.025* |
| | (0.008) | (0.014) | (0.010) | (0.011) | (0.009) | (0.026) | (0.010) | (0.020) | (0.014) |
| Constant | 3.063* | -0.705 | 3.482** | 3.224* | 4.438*** | -4.228 | 3.578** | 1.621 | 2.913 |
| | (1.701) | (3.123) | (1.462) | (1.845) | (1.406) | (6.567) | (1.623) | (2.324) | (2.937) |
| Observations | 246 | 129 | 117 | 197 | 219 | 62 | 108 | 76 | 132 |
| Number of countries | 51 | 23 | 28 | 39 | 45 | 15 | 28 | 12 | 32 |
| Number of programs | 84 | 45 | 39 | 67 | 75 | 22 | 36 | 26 | 44 |
| R-squared | 0.396 | 0.384 | 0.439 | 0.412 | 0.485 | 0.552 | 0.496 | 0.603 | 0.488 |
| WEO vintage FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Tax-to-GDP Forecast Errors Over-Optimism | 0.597 | 0.027 | 1.226 | 0.397 | 0.583 | 0.282 | 1.302 | -0.146 | 0.172 |
| Over-Optimism (p-value) | 0.008 | 0.922 | 0.000 | 0.095 | 0.017 | 0.447 | 0.000 | 0.657 | 0.539 |

Note: OLS Estimates. Dependent variable is tax revenue-to-GDP forecast errors for the combined horizons covering the first three years following program approval (h=1, 2, 3). In all specifications, tax policy yields and the index measuring the implementation and depth of revenue-related structural reforms are interacted. Columns (2) to (9) focus on different samples. Column (2) includes only Emerging Markets, columns (3) includes Low-income and Developing Countries, columns (4) includes non-Fragile and Conflict-Affected Countries, columns (5) includes non-fuel exporters, columns (6) includes GRA programs, columns (7) includes PRGT programs, columns (8) includes Non-Disbursing programs, and columns (9) includes programs with all completed reviews. Robust standard errors clustered by country are shown in parentheses. The use of *** denotes a coefficient significantly different from zero at the 0.01 confidence level; ** 0.05; * 0.10. GRA-PRGT classifications are based on the resource account from which Fund-supported programs draw, while the EM–LIDC classification follows the IMF World Economic Outlook (WEO). The two classifications overlap but are not identical: some PRGT eligible countries are classified as emerging markets (EMs) under the WEO classification, and some countries classified as LIDCs in the WEO are not PRGT eligible. Direct links between the LIDC grouping and PRGT supported programs should therefore be avoided.

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